String Theory

Leonard Susskind

Supplementary track, 2010 Fall String Theory And M Theory
Transcript-derived notes curated by LazyingArt LLC with Video2Book

Original lectures by Leonard Susskind. Transcript-derived course notes curated by LazyingArt LLC with Video2Book.

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Chapter 1

String Theory and M-Theory

These notes follow Leonard Susskind's opening lecture on string theory, with curation by LazyingArt LLC. The lecture begins where string theory actually began: not with quantum gravity, but with hadron physics. We first follow an experimental pattern, then a scattering puzzle, then a geometric re-description, and only after that do we turn to the mathematics of strings. The order is essential. The lecture is not presenting a finished formalism and then sprinkling in some history; it is letting the need for the formalism emerge.

1.1 Hadrons, Towers, and Regge Trajectories

String theory entered physics through hadrons. In the late 1960s one did not yet speak in the settled modern language of quarks and gluons, but one did have spectroscopy. Protons, neutrons, mesons, and their excitations did not appear as isolated accidents. They came in families, and the families showed a remarkably simple pattern.

The organizing device was the Chew–Frautschi plot: angular momentum vertically, mass squared horizontally. Susskind moves rather freely between angular momentum, spin, and the symbol L; for the written notes it is cleaner to denote the vertical axis by J. The experimental fact was that hadronic families lay approximately on straight lines,

$$J \propto M^2, \tag{1.1}$$

and, more strikingly, different families had approximately the same slope. In this sense there was something like a universal Regge slope. One may summarize the observed quantization schematically as

$$\Delta J = 1 \qquad \Longrightarrow \qquad \Delta M^2 \approx \text{constant}. \tag{1.2}$$

For mesons the angular momenta are integers; for baryons they are half-integers. But in both cases the same broad phenomenon appears: higher and higher spin states, marching upward in nearly parallel lines on the J versus M^2 plane.

This is already a strong clue. One can spin up a basketball or an atom, but the resulting trajectory is not a straight line in M^2, and in any case the thing eventually breaks apart or ionizes. The hadronic pattern is too simple, too linear, and too persistent. It tells us that hadrons are not point particles. They are composite objects that can be excited rotationally, and they must have a rather special kind of stretchability.

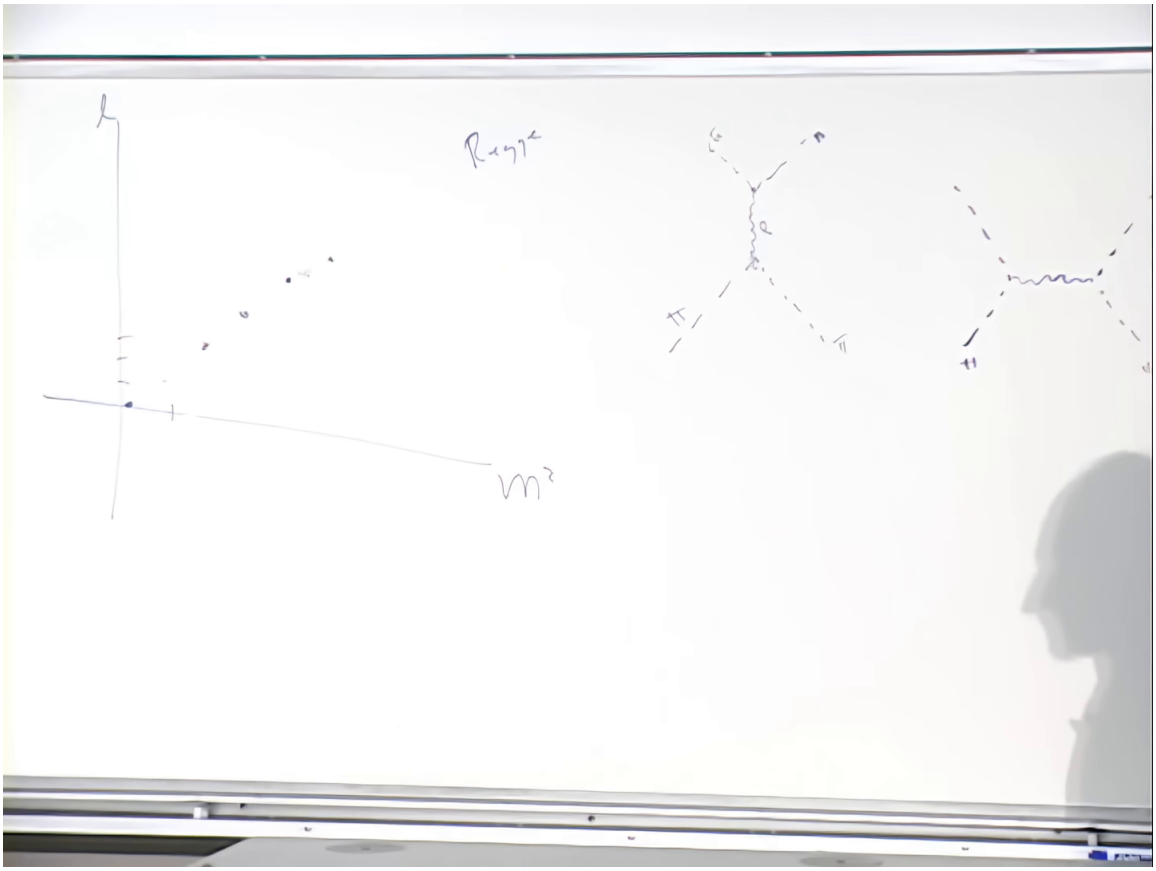

Figure 1.1: Regge trajectory and dual scattering sketches.

This is the first beat of the lecture. It is historical, but not decorative. The data are already suggesting the right sort of object: something extended, excitable, and apparently not eager to give out as the spin increases.

1.2 Meson Scattering and the Duality Tension

That was one observation. Then there was another. Susskind now pivots from spectroscopy to scattering, and he does it concretely: take pion–pion scattering. In conventional field-theoretic language, two pions may come together, form a ρ-like resonance, and then emerge again as two pions. But if that diagram exists, then the crossed diagram exists as well: the same ingredients also describe the exchange of a ρ-like state between the pions.

So far there is no mystery. The mystery arrives when we remember that the ρ is not a single isolated particle. It sits on a whole Regge trajectory. Therefore in the annihilation channel one should not insert only the ρ, but also the ρ', the ρ'', and so on. Likewise in the exchange channel. The lecture never turns this into a detailed amplitude analysis, but the conceptual statement can be written schematically as

$$\mathcal{A}_{\pi\pi}(s,t) \sim \sum_n \mathcal{A}^{(s)}_{\rho_n}(s) \qquad \text{and} \qquad \mathcal{A}_{\pi\pi}(s,t) \sim \sum_n \mathcal{A}^{(t)}_{\rho_n}(t). \tag{1.3}$$

The point is not that these are exact formulas in the lecture. The point is that when people

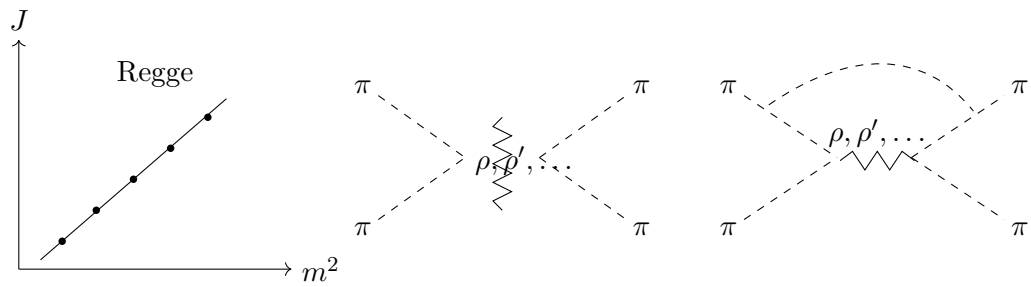

Figure 1.2: A clean redraw of the board cluster: a Regge plot at left and the two channel pictures at right.

numerically summed the known resonances in one channel, they seemed to be getting, approximately and already, what the other channel would have contributed.

That is the duality tension. Ordinary quantum field theory says: if this diagram exists and that diagram exists, then we must add them. The hadronic data seemed instead to be saying that the sum over a full tower in one channel was already reproducing the other. Naively adding both looked like overcounting.

The board itself makes this transition visible: the Regge plot and the two channel sketches are drawn as one conceptual cluster, as if the spectral regularity and the dual-channel puzzle were already different faces of the same thing.

1.3 From Quark Pictures to World-Sheets

At this point the lecture becomes exploratory in exactly the right way. Susskind does not say: here is the formal object and here is its theorem. He says, in effect, something is going on here, and the Feynman-diagram bookkeeping is no longer the right way to see it.

The next move is to draw quark diagrams with a topological character. Imagine quark and antiquark lines running upward in time. If we view the whole history as a two-dimensional sheet, then one slice through the sheet looks like annihilation and recreation, while another slice looks like exchange. The two channel pictures are no longer two separate dynamical mechanisms; they are two ways of cutting one underlying history.

This is where the jargon arrives:

Definition 1.1. A world sheet is the two-dimensional spacetime history traced by a string, just as a world line is the one-dimensional spacetime history traced by a point particle.

The lecture is careful here. It does not claim that a full string theory has been derived. It claims something more primitive and, historically, more accurate: a two-dimensional object can naturally account for the duality between the channels.

1.3.1 Question & Answer

Why think of a string-like bridge between the quark and antiquark rather than an ordinary exchanged particle?

Because the object in question is not localized at a single intermediate point. If we take an instant of time, we do not see two endpoints and one exchanged quantum. We see a quark, an antiquark, and a whole bunch of stuff in between. Susskind explicitly leaves open whether that intervening stuff is truly continuous or only an approximation to continuity; quantum mechanics makes the distinction subtle. But whatever it is, it fills the interval between the endpoints. Moreover, the Regge trajectories showed no sign of ending. A shoelace with two golf balls at its ends eventually breaks when spun hard enough. The hadronic object, as far as the data then showed, did not. That is what made the one-dimensional extended picture feel more basic than ordinary particle exchange.

Once one says this, the first hadronic clue returns in a new form. A hadron that is a string with two quarks attached to it is exactly the kind of object one can spin up, deform, and analyze energetically.

1.4 Flux Tubes, Stretchability, and the Historical Turn

The lecture now pauses the crude historical picture and reinterprets it in modern terms. Today we would say that the string between a quark and an antiquark is a flux tube of the gluon field. The most useful contrast is with electrodynamics. Between opposite electric charges the field lines spread out, so the field weakens as the charges are pulled apart. That is not a string.

In QCD the nonlinearities of the gauge field change the story. The effective field lines do not simply spread; they are drawn into a tube. As the endpoints separate, the tube gets longer rather than thinner. One may think of the tube as made of gluonic structure. Then the key property is that the energy of stretching does not merely increase the distance between fixed constituents; it goes into creating more constituents in between, so that the density along the tube stays roughly fixed.

This is why Susskind leans on the analogy with a rubber band whose number of molecules is *not* fixed. A normal rubber band stretches by separating the same molecules further and further until it breaks. The hadronic string, in the lecture's modernized picture, stretches by producing more stuff in the gaps. That is why it can remain string-like over a long range of excitation.

The same physical lesson appears in the superconductor analogy. A confined line of magnetic flux in a superconductor has approximately uniform field strength along its length. Uniform field strength means uniform energy density, and uniform energy density means energy proportional to length. We do not need the full superconducting construction here; the moral is enough. Field theory really does produce string-like objects with tension.

Only after making this point does Susskind step back historically. The mathematics that grew out of these hadronic ideas did not, in its simplest form, become the final theory of hadrons. It kept producing, embarrassingly, a massless spin-two particle. What had looked like a nuisance in hadron physics would later be recognized as the graviton. So the mathematical theory born from hadron phenomenology turns out to look more like quantum gravity than like hadronic spectroscopy. But before one gets to that reinterpretation, one still needs the basic mechanics of the string itself, and for that the lecture turns sharply to kinematics.

1.5 Relativity Reorganized in the Infinite-Momentum Frame

Before talking about string mathematics, the lecture talks about relativity versus nonrelativistic kinematics. This is not a digression. It is the license for everything that follows.

For a slow particle the energy is

$$E_{\mathrm{nr}} = \frac{p^2}{2m} + B, \tag{1.4}$$

where B is an additive internal or binding term that does not depend on the state of motion. In relativity the exact energy is

$$E = \sqrt{p^2c^2 + m^2c^4}. \tag{1.5}$$

If $p \ll mc$, then we can factor out the large term and expand:

$$E = mc^2\sqrt{1 + \frac{p^2}{m^2c^2}} \tag{1.6}$$

$$\approx mc^2 + \frac{p^2}{2m}. \tag{1.7}$$

This is the familiar nonrelativistic limit. But Susskind insists that it is only an approximation, and it is a good approximation only when all relevant particles are moving slowly, not merely the center of mass of the system.

Now comes the other expansion, the one that matters for strings. Boost the entire system so that every constituent carries a very large positive momentum along the z-axis. From here on we set $c = 1$. Then

$$E_i = \sqrt{P_{zi}^2 + p_{\perp i}^2 + m_i^2}, \qquad p_{\perp i}^2 = p_{xi}^2 + p_{yi}^2. \tag{1.8}$$

The board notation shifts between P_3 and P_z; we standardize to P_z. Since P_{zi} is large while $p_{\perp i}$ and m_i remain fixed, we expand in powers of $1/P_{zi}$:

$$E_i = P_{zi}\sqrt{1 + \frac{p_{\perp i}^2 + m_i^2}{P_{zi}^2}} \tag{1.9}$$

$$\approx P_{zi} + \frac{p_{\perp i}^2 + m_i^2}{2P_{zi}}. \tag{1.10}$$

Summing over all constituents gives

$$E = \sum_i P_{zi} + \sum_i \frac{p_{\perp i}^2}{2P_{zi}} + \sum_i \frac{m_i^2}{2P_{zi}}. \tag{1.11}$$

The first term is just the total momentum down the z-axis. It is huge, but it is also conserved. As an additive constant in the energy it may be dropped without affecting the dynamics. The remaining expression can be read as a Hamiltonian:

$$H = \sum_i \left(\frac{p_{\perp i}^2}{2P_{zi}} + \frac{m_i^2}{2P_{zi}} \right). \tag{1.12}$$

The board then reinterprets the same formula as the generator of time evolution,

$$i\,\partial_t = H, \tag{1.13}$$

with $\hbar = 1$.

Now the lecture makes the crucial interpretation. In this transverse description the quantity in the denominator plays the role that mass usually plays in nonrelativistic mechanics. So P_z acts

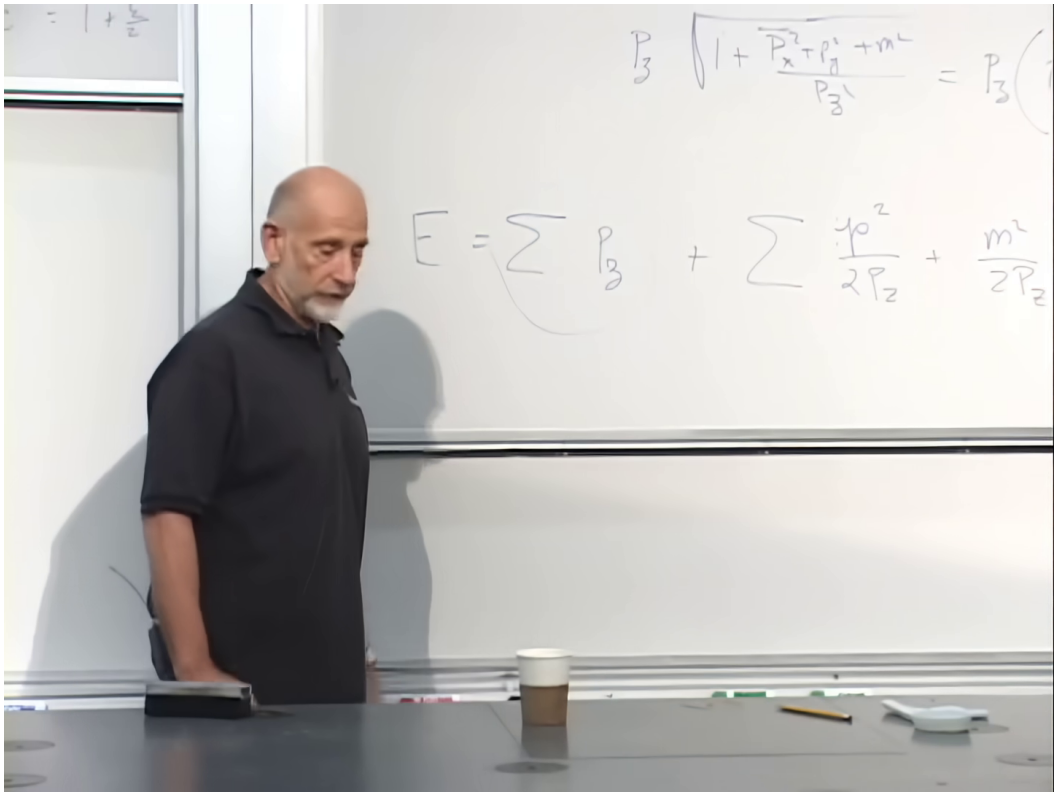

Figure 1.3: Large-P_z expansion of the relativistic energy.

as transverse inertia. Meanwhile the term involving m_i^2 is independent of the transverse state of motion, and so it plays the role of internal or binding energy. This is the exact structural reason that a relativistic system, viewed in the infinite-momentum frame, behaves nonrelativistically in the transverse plane.

Susskind then adds the physical explanation. Because the Hamiltonian scales like $1/P_z$, the evolution becomes slow when P_z is large. That is just time dilation. One is not denying relativistic physics; one is reorganizing it into a form that is calculable.

1.6 Open Strings as Masses and Springs

By what chutzpah do we now use nonrelativistic physics to describe a relativistic string? By the chutzpah just earned. In the infinite-momentum frame the transverse dynamics really are nonrelativistic, and that is enough to write down the simplest model.

Take an open string. Introduce a coordinate σ along it, with

$$0 \leq \sigma \leq \pi.$$

The choice of π is conventional; Susskind chooses it because later closed strings will naturally run from 0 to 2π.

Start discretely. Think of the string as a chain of point masses joined by springs. If $X_i(t)$ denotes

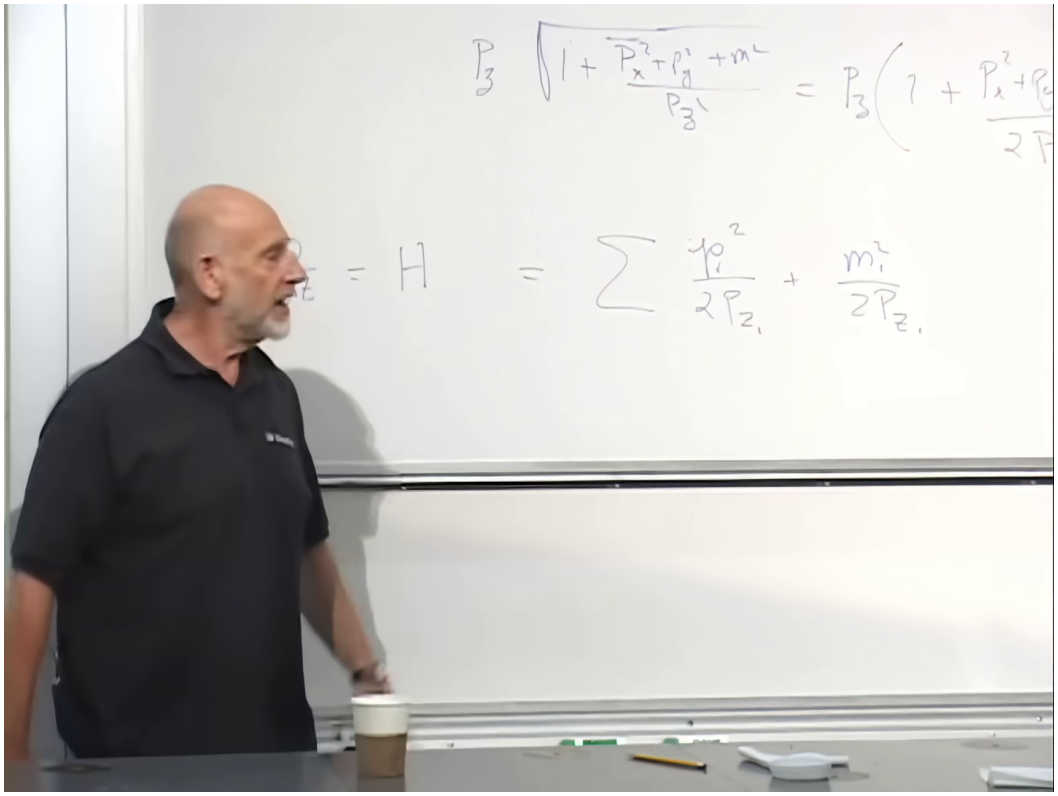

Figure 1.4: Hamiltonian form of the light-cone energy.

the transverse position of the i-th mass point, then the energy is

$$E = \sum_i \left(\frac{\mu}{2} \dot{X}_i^2 + \frac{k}{2} (X_i - X_{i+1})^2 \right). \tag{1.14}$$

Here $\dot{X}_i^2$ abbreviates the sum of the two transverse components, and the second term is simply Hooke's law.

The lecture is careful about the continuum limit. As the beads become denser, the mass of each bead must go to zero so that the total mass remains finite. At the same time the spring constant must scale upward: stretching a tiny segment by a fixed amount is harder than stretching a long segment by the same amount. When the spacing goes to zero, the sum becomes an integral and the nearest-neighbor difference becomes a derivative:

$$\sum_i \longrightarrow \int_0^\pi d\sigma, \qquad X_i - X_{i+1} \longrightarrow \partial_\sigma X. \tag{1.15}$$

Thus the continuum energy takes the form

$$E \sim \int_0^\pi d\sigma \left[\frac{1}{2} \dot{X}^2(\sigma, t) + (\partial_\sigma X(\sigma, t))^2 \right]. \tag{1.16}$$

The exact coefficients are not fully fixed by the board, and Susskind explicitly says he has absorbed constants by convention. What matters is the structure: kinetic energy from $\partial_t X$, potential energy from the stretching measured by $\partial_\sigma X$.

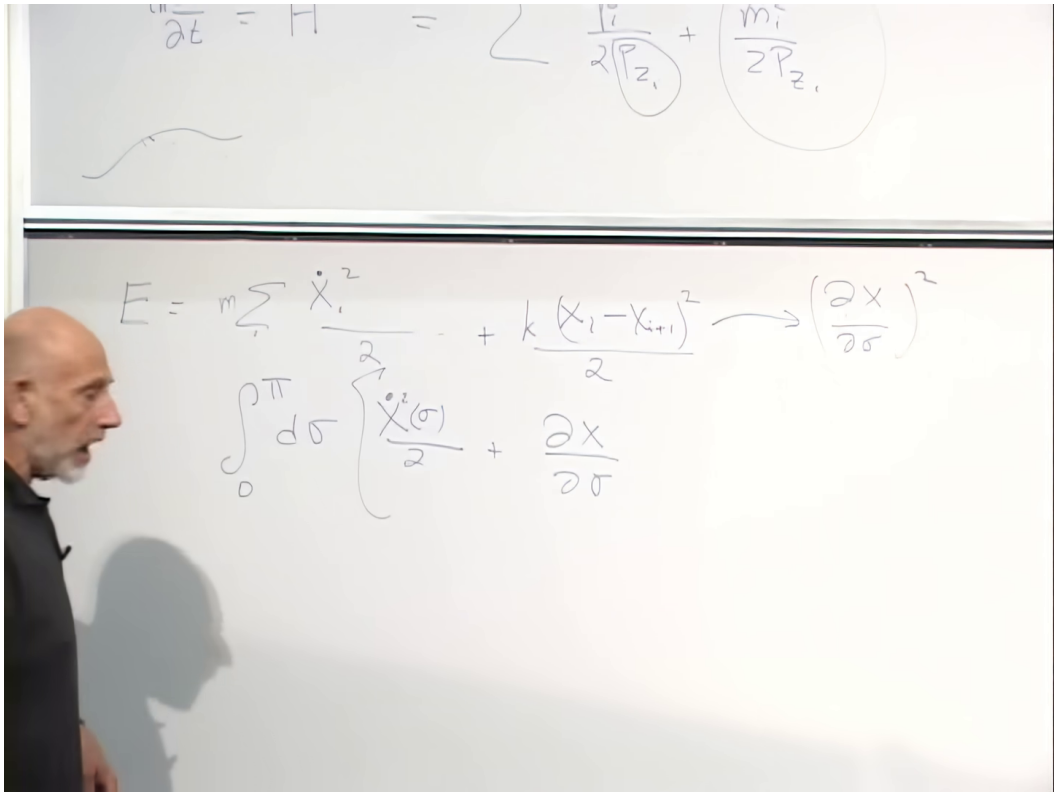

Figure 1.5: Discrete string energy and continuum σ-derivative limit.

The lecture pauses here for a useful distinction. If we were writing the Lagrangian, the potential term would come with a minus sign. But Susskind wants the energy, the Hamiltonian, so the terms appear with plus signs. That is the object we will need when we ask what the internal motion of the string contributes to the observable particle.

1.7 Internal Energy as M^2

We now return to the key interpretive question. The string may vibrate and stretch, but as a whole it is still one object. We may call it a particle if we like. Its center of mass describes the overall motion of the particle, while the relative stretching and internal vibration describe its internal energy.

What, then, does the Hamiltonian of the previous section correspond to? In ordinary rest-frame reasoning one would identify internal energy with mass. But the whole point of the infinite-momentum analysis was that in this frame internal energy enters as mass squared. Thus the string Hamiltonian should be read as contributing to M^2, not to M:

$$H_{\text{int}} \propto M^2. \tag{1.17}$$

Susskind is quite emphatic about this. It is not just an analogy; it is the precise way relativity reorganizes itself in the light-cone frame.

This immediately explains why string excitations naturally come in evenly spaced units of M^2. The string is a collection of coupled oscillators. We do not quantize it fully in this lecture, but

the principle is already clear: oscillator energies are quantized in equal steps, so string excitations contribute equal steps to the internal Hamiltonian, hence to mass squared. This is exactly the kind of spacing the Regge trajectories seemed to demand.

1.7.1 Question & Answer

Why does the same string look Hooke-like in one frame but tension-like in its rest frame?

Because the Hooke-like energy is the effective internal energy of the boosted description, and that quantity maps to M^2, not directly to the rest energy M. The rest-frame string is recovered only after we translate the light-cone result back into ordinary relativistic language.

Let us go through the argument the way the lecture does. Consider a uniformly stretched string of physical length L. Along the parameter interval $0 \leq \sigma \leq \pi$ we then have, schematically,

$$\partial_\sigma X \sim \frac{L}{\pi}. \tag{1.18}$$

Insert this into the stretching term:

$$H_{\text{stretch}} \sim \int_0^\pi d\sigma \, (\partial_\sigma X)^2 \tag{1.19}$$

$$\sim \int_0^\pi d\sigma \left(\frac{L}{\pi}\right)^2 \tag{1.20}$$

$$= \pi \left(\frac{L}{\pi}\right)^2 \propto L^2. \tag{1.21}$$

So in the effective nonrelativistic description the stretching energy behaves like Hooke's law:

$$H_{\text{stretch}} \propto L^2. \tag{1.22}$$

But this energy is to be identified with M^2. Therefore

$$M^2 \propto L^2, \qquad \text{so} \qquad M \propto L. \tag{1.23}$$

In the rest frame the energy of the string is its mass, and so

$$E_{\text{rest}} = M = T\,L. \tag{1.24}$$

Thus the rest-frame string is not an ordinary Hooke-law spring at all. It is a constant-tension object, precisely of the flux-tube type discussed earlier.

Susskind is careful here: this is an indirect deduction. We are not claiming to know how an arbitrary ordinary rubber band would behave if it were driven relativistically. What we have done is use the frame in which the transverse dynamics become simple, derive the L^2 behavior there, and then translate back. That is why the result is trustworthy even though the route is indirect.

1.7.2 After-Break Clarifications

After the break the lecture revisits the same point more slowly, because this is where the sleight of hand can be missed. The same relativistic formula

$$E = \sqrt{p^2 + m^2} \tag{1.25}$$

admits two different expansions.

If p is small and m is large, then

$$E \approx m + \frac{p^2}{2m}. \tag{1.26}$$

This is the ordinary rest-frame logic: internal energy is read as mass. If, on the other hand, p is very large, then

$$E \approx p + \frac{m^2}{2p}. \tag{1.27}$$

After the conserved momentum term is removed, the remaining energy is proportional to m^2. That is the whole conceptual switch. In one frame the relevant internal quantity is M; in the boosted frame it is M^2.

The lecture also addresses a natural question about residual motion along the boost direction. In light-cone string theory one does not introduce independent longitudinal degrees of freedom in the same way one introduces the transverse ones. The description is transverse, and yet it remains Lorentz invariant. In the lecture this is presented as a remarkable fact rather than something to prove immediately, and that is the right level to preserve here.

Finally, Susskind returns to the electron question. Why do hadrons show low-lying higher-spin excitations while electrons do not? The answer is the rotational estimate

$$E_{\text{rot}} \sim \frac{L^2}{2I}, \qquad I \sim MR^2. \tag{1.28}$$

Since angular momentum comes in quanta, the first excited rotational state costs roughly

$$\Delta E \sim \frac{\hbar^2}{2I}. \tag{1.29}$$

If the object is tiny, then R is tiny, hence I is tiny, and the excitation energy is enormous. Electrons are experimentally exceedingly small, so any such excitations would lie far beyond accessible collision energies. Hadrons are large on the scale of their own internal dynamics, so their moments of inertia are large enough that spinning them up by one unit costs only hadronic energies.

This closes the loop back to the opening observation. The Regge plots were not merely classification devices. They were telling us that hadrons behave as extended objects with accessible rotational and vibrational excitations.

1.8 Summary

The lecture begins with hadron phenomenology and lets string theory emerge from it. Straight, parallel Regge trajectories suggest that hadrons are composite, stretchable objects rather than points. The duality puzzle in pion scattering then suggests that what field theory wants to count as separate channels may really be two slicings of one underlying two-dimensional history, a world sheet.

From there the lecture pauses to reinterpret the old picture in modern QCD terms: a flux tube has constant energy per unit length, and so it behaves like a string with tension. Only then does the mathematical turn arrive. In the infinite-momentum frame a relativistic system becomes nonrelativistic in the transverse plane, with P_z playing the role of inertia and the internal term

behaving like mass squared. That allows us to model an open string as masses and springs, pass to the continuum Hamiltonian built from $\dot{X}^2$ and $(\partial_\sigma X)^2$, and then read that Hamiltonian as contributing to M^2. The payoff is immediate: Hooke-law stretching in the light-cone description becomes linear-tension behavior in the rest frame, and string excitations naturally organize themselves in evenly spaced quanta of mass squared.

Chapter 2

Lecture 2: String Theory and M-Theory

These notes follow Leonard Susskind's second lecture on string theory, curated in this archive by LazyingArt LLC. The lecture proceeds in a very deliberate order. First it stocks the blackboard with the interval calculus and Fourier facts that would otherwise interrupt the story later. Then it pauses over a conceptual question that matters more than it first appears to: in what sense can a string, made of many pieces, still count as a particle? Only after that does it return to the infinite-momentum or light-front point of view, build the open string as a chain of masses and springs, take the continuum limit, solve the endpoint puzzle, and quantize the resulting modes.

2.1 Mathematical Preliminaries on the Interval $[0, \pi]$

Susskind begins by saying explicitly why he is doing this: if these preliminaries were not written down now, they would have to be inserted later in the middle of the string theory, and that would be a nuisance. So we begin with the reservoir of tools.

Take a function $x(\sigma)$ on the interval $0 \leq \sigma \leq \pi$. The choice of letters is not accidental. He uses x and σ because these are the notations string theory will keep returning to. The interval $[0, \pi]$ is likewise chosen in advance because it is the natural interval for the open string in this lecture.

We approximate the continuous function by values on a lattice of N points. If

$$\Delta\sigma = \frac{\pi}{N}, \tag{2.1}$$

then the standard replacements are

$$x(\sigma) \longrightarrow x_i, \tag{2.2}$$

$$\Delta x_i \equiv x_i - x_{i-1} \approx \frac{\partial x}{\partial \sigma}\,\Delta\sigma, \tag{2.3}$$

$$\sum_i x_i\,\Delta\sigma \longrightarrow \int_0^\pi x(\sigma)\,d\sigma. \tag{2.4}$$

This is just calculus, but it is precisely the calculus we will soon use for the string.

The next distinction is between endpoint behaviors. Here the lecture is still mathematical rather than physical, but the physical pictures are already visible.

Dirichlet boundary conditions. These fix the value of the function at the endpoints:

$$x(0) = x(\pi) = 0. \tag{2.5}$$

A violin string with its ends pinned down is the elementary model.

Neumann boundary conditions. These instead fix the slope at the endpoints:

$$\left.\frac{\partial x}{\partial \sigma}\right|_{\sigma=0} = \left.\frac{\partial x}{\partial \sigma}\right|_{\sigma=\pi} = 0. \tag{2.6}$$

The function need not vanish there; it is only required to be flat.

The lecture briefly wanders through organ pipes and other examples, but the moral is simple. Dirichlet means fixed endpoints. Neumann means free endpoints. Susskind even gives a mechanical picture of a string whose ends pass through very light rings sliding on poles: not held down, but free to move. That is the kind of system for which Neumann conditions are natural.

There is also a small caution here. Piecewise continuous functions are allowed in Fourier analysis, but for the purposes of string theory we do not want jumps in the string itself: a jump would mean the string had broken.

Once the endpoint behavior is fixed, the Fourier basis is fixed with it. On $[0, \pi]$ one writes

$$x_D(\sigma) = \sum_{n=1}^{\infty} x_n \sin n\sigma, \tag{2.7}$$

$$x_N(\sigma) = \sum_{n=0}^{\infty} x_n \cos n\sigma. \tag{2.8}$$

Sines vanish at the endpoints, which is why they go with Dirichlet. Cosines have zero derivative at the endpoints, which is why they go with Neumann. The lecture also emphasizes a fact we will use later without apology: differentiation swaps the two classes. The derivative of a Dirichlet function is Neumann; the derivative of a Neumann function is Dirichlet.

To make these expansions useful, we need the orthogonality integrals. In the form relevant later,

$$\int_0^{\pi} \cos n\sigma \cos m\sigma \, d\sigma = 0 \qquad (n \neq m), \tag{2.9}$$

$$\int_0^{\pi} \cos^2 n\sigma \, d\sigma = \frac{\pi}{2} \qquad (n \geq 1), \tag{2.10}$$

$$\int_0^{\pi} d\sigma = \pi, \tag{2.11}$$

$$\int_0^{\pi} \sin n\sigma \sin m\sigma \, d\sigma = \frac{\pi}{2}\delta_{nm} \qquad (n, m \geq 1). \tag{2.12}$$

Remark 2.1. The transcript becomes garbled around the exceptional $n = 0$ cosine mode, but the intended mathematics is clear. The nonzero cosine modes have norm $\pi/2$, while the constant mode $\cos 0\sigma = 1$ integrates to π.

The last item placed on the board is the harmonic oscillator. After rescaling the coordinate so that the mass is one, the energy reads

$$E_{\text{osc}} = \frac{1}{2}\dot{x}^{\,2} + \frac{1}{2}\omega^2 x^2, \tag{2.13}$$

and the corresponding Lagrangian is

$$L_{\text{osc}} = \frac{1}{2}\dot{x}^{\,2} - \frac{1}{2}\omega^2 x^2. \tag{2.14}$$

The lecture contains a verbal slip here, but the mathematics is standard: the Lagrangian is $T - V$, not "kinetic energy squared minus potential energy squared." The quantum fact we keep in reserve is that oscillator levels come in steps of ω, up to the overall ground-state offset.

2.2 What Makes a String a Particle?

Now the lecture deliberately interrupts its own mathematical flow. Before building the string, it asks a question that could otherwise become a source of confusion for the rest of the course: what do we mean by a particle at all?

A string, after all, is about to be modeled as a large number of pieces connected by springs. If that is so, why should it behave like a particle rather than like an ordinary extended object?

The first part of the answer is negative. Pointlikeness is not the criterion. Electrons are surrounded by virtual clouds. Protons and neutrons are obviously composite. The lecture insists that what we call a particle need not be literally structureless.

Nor is it enough merely to have a mass and a center-of-mass position. A box of gas has those. A cup of coffee has those. Yet no physicist would naturally call a cup of coffee a particle.

2.2.1 Question & Answer

Question. When does a composite object count as a particle rather than "mush"?

Answer. A composite object counts as a particle when the excitation spectrum above its ground state is discrete and well separated. If there are large enough gaps to the first few excited states, the object behaves as an isolated quantum species. If instead there are innumerable nearby states at arbitrarily small separations, the object behaves like continuum matter.

This is the lesson of the lecture's coffee example. A cup of coffee can be excited by tiny disturbances: one can stir it slightly, excite a sound mode, disturb a molecule, or create a phonon with an absurdly small amount of energy. The nearby states are so dense that they practically form a continuum. That is mush.

A proton is different. It certainly has excitations, but it takes a substantial amount of energy to get to them. The nearby spectrum is not continuous. An electron is more isolated still. One does not have a ladder of electron states lying immediately above the electron.

So the criterion becomes spectral rather than ontological. The question for string theory is not whether the string is made of parts. Of course it is. The question is whether the internal excitations are separated by large enough gaps to make the lowest state behave as a particle. In the version of string theory under discussion, the gap is taken to be enormous, on the scale of the Planck mass, and that is why the string may still present itself as a particle in ordinary experiments.

This detour is not philosophical ornament. It prepares the central claim of the lecture's last third: once the string is quantized, its spectrum comes out discrete enough to justify calling the resulting states particles.

2.3 Review of the Infinite-Momentum / Light-Front Trick

Only after that conceptual pause does Susskind return to the framework from the previous lecture. The question is this: how can an extended relativistic system ever be described by something as simple as nonrelativistic mechanics?

He first emphasizes what does *not* work. A rapidly spinning baton is not well described by nonrelativistic physics just because we happen to sit in the frame where its center of mass is simple. The issue is the relative motion of its parts. If the ends are moving relativistically relative to each other, then the ordinary nonrelativistic approximation fails.

The trick is different. We boost the whole system to a very large momentum along some chosen axis, say the z-axis, and then examine only the motion in the transverse plane. In that description the transverse dynamics become exactly, not approximately, nonrelativistic in form.

Start from the relativistic energy

$$E = \sqrt{P_z^2 + p_x^2 + p_y^2 + M^2}, \tag{2.15}$$

where M is the rest mass of the full system and P_z is the large longitudinal momentum. When P_z is the dominant scale, we expand:

$$E \approx P_z + \frac{p_x^2 + p_y^2 + M^2}{2P_z}. \tag{2.16}$$

The first term is the trivial boost energy. It depends on the conserved longitudinal momentum but not on the internal state, so for the questions at hand we subtract it and write

$$E - P_z = \frac{p_x^2 + p_y^2 + M^2}{2P_z}. \tag{2.17}$$

The lecture then makes the crucial interpretive move. Energy is Hamiltonian. Hamiltonian governs time evolution. But if P_z becomes very large, the right-hand side tends to zero, which means that in the ordinary time variable the internal motion appears to freeze. This is just time dilation.

We do not want to throw away the internal motion. We want to follow it. So we rescale time. Equivalently, up to a conventional overall normalization, we multiply by P_z. Then the expression has a finite limit:

$$P_z(E - P_z) = \frac{1}{2}\left(p_x^2 + p_y^2 + M^2\right). \tag{2.18}$$

This now looks like a two-dimensional nonrelativistic Hamiltonian: transverse momentum squared divided by two, plus an internal contribution. The only conceptual twist is that the internal contribution is proportional to M^2, not M.

On the board, the line beginning with $H =$ records exactly this reinterpretation. The board itself does not finish writing $H = i\,d/dt$, but the lecture makes that identification in words: Hamiltonian is what governs time evolution.

Remark 2.2. Susskind briefly complains about the terminology "light cone" and says that "light front" would be better. We keep only the mathematical point. After the large boost and the time rescaling, the transverse dynamics are exactly nonrelativistic in form.

There is one more feature he flags, without dwelling on it. For strings, the longitudinal internal motion is constrained by the transverse motion. That is part of what makes the method so powerful. In the present lecture, however, this remains background motivation; the active calculation now moves to the transverse coordinates alone.

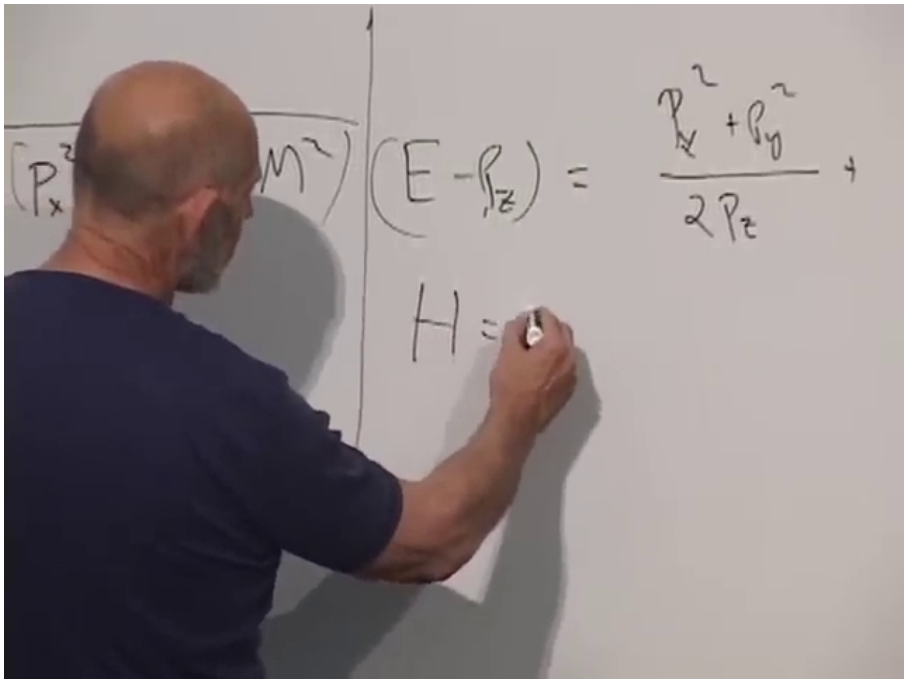

Figure 2.1: Hamiltonian form of the boosted energy. The board line beginning with $H =$ marks the shift from the boosted-energy expansion to its Hamiltonian interpretation.

2.4 A String as Mass Points and Springs

Now we come to strings themselves. In the boosted description the string moves only in two transverse directions, which we call X and Y. The lecture begins with the discrete model: N mass points joined by $N-1$ springs.

The coordinate σ labels position along the string and runs from 0 to π. This is not forced; it is chosen. Susskind even remarks that one could equally well choose 0 to 1 or 0 to 2. The reason for choosing $[0, \pi]$ is pedagogical. He wants to reserve $[0, 2\pi]$ for the closed string later on.

The discrete energy of the open string is

$$E = \sum_i \left[\frac{\mu}{2}(\dot{X}_i^2 + \dot{Y}_i^2) + \frac{k}{2}((\Delta X_i)^2 + (\Delta Y_i)^2) \right]. \tag{2.19}$$

The board writes the bead mass as m, but the lecture itself later corrects this and introduces μ as the nonrelativistic analog mass of each bead, so we use μ from the start in the cleaned notes. Likewise, the spring constant is called both k and κ in speech; we write k.

When Susskind writes the formula first for X, he immediately says: wherever I write X, you must also add the same thing with Y. That small spoken reminder matters. The string lives in two transverse directions, not one.

The next task is to choose the large-N scaling correctly. If the number of beads grows while each bead keeps a fixed mass, then the total mass diverges. That is not the continuum limit we want. We therefore choose

$$\mu = \frac{1}{N}, \tag{2.20}$$

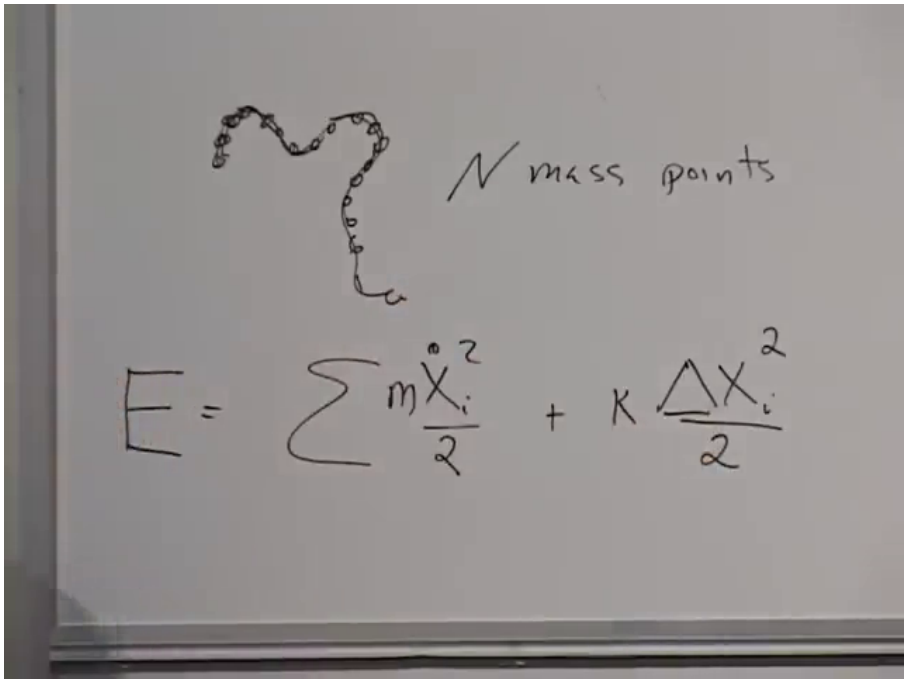

Figure 2.2: Discrete string as mass points and springs. The lecture moves directly from the bead-chain sketch to the discrete energy formula.

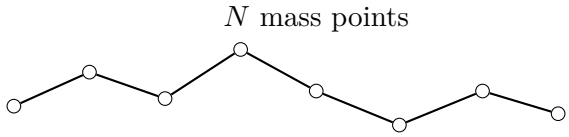

Figure 2.3: A clean redraw of the bead-chain model used to take the continuum limit.

so that the total analog mass remains fixed.

The spring constant must scale in the opposite direction. Springs in series make a long chain easier to stretch, so if the overall stiffness of the string is to stay finite, the microscopic springs must become stiffer as the chain is refined. Susskind chooses

$$k = \frac{N}{\pi^2}. \tag{2.21}$$

The factor of π^2 is a convention inserted only to simplify later formulas. The essential point is the $k \sim N$ scaling.

This is the real payoff of the opening calculus detour. We now have a discrete mechanical chain and a precise prescription for taking its continuum limit.

2.5 Continuum Energy, Lagrangian, and Wave Interpretation

At this point the lecture uses, almost verbatim, the formulas it put on the board at the beginning. We have $\Delta\sigma = \pi/N$, we have sums, and we have nearest-neighbor differences. Now we turn them into integrals and derivatives.

A worked derivation of the continuum limit. For the X-sector kinetic term,

$$\sum_i \frac{\mu}{2}\dot{X}_i^2 = \sum_i \frac{1}{2N}\dot{X}_i^2 \tag{2.22}$$

$$= \frac{1}{2\pi}\sum_i \dot{X}_i^2\,\Delta\sigma \tag{2.23}$$

$$\longrightarrow \frac{1}{2\pi}\int_0^\pi d\sigma \left(\frac{\partial X}{\partial \tau}\right)^2. \tag{2.24}$$

For the spring term we use the continuum replacement

$$\Delta X_i \approx \frac{\partial X}{\partial \sigma}\,\Delta\sigma. \tag{2.25}$$

Then

$$\sum_i \frac{k}{2}(\Delta X_i)^2 \approx \sum_i \frac{N}{2\pi^2}\left(\frac{\partial X}{\partial \sigma}\right)^2 (\Delta\sigma)^2 \tag{2.26}$$

$$= \sum_i \frac{1}{2N}\left(\frac{\partial X}{\partial \sigma}\right)^2 \tag{2.27}$$

$$= \frac{1}{2\pi}\sum_i \left(\frac{\partial X}{\partial \sigma}\right)^2 \Delta\sigma \tag{2.28}$$

$$\longrightarrow \frac{1}{2\pi}\int_0^\pi d\sigma \left(\frac{\partial X}{\partial \sigma}\right)^2. \tag{2.29}$$

Exactly the same steps apply to Y.

So the continuum energy becomes

$$E = \frac{1}{2\pi}\int_0^\pi d\sigma \left[\left(\frac{\partial X}{\partial \tau}\right)^2 + \left(\frac{\partial Y}{\partial \tau}\right)^2 + \left(\frac{\partial X}{\partial \sigma}\right)^2 + \left(\frac{\partial Y}{\partial \sigma}\right)^2\right], \tag{2.30}$$

and the continuum Lagrangian is

$$L = \frac{1}{2\pi}\int_0^\pi d\sigma \left[\left(\frac{\partial X}{\partial \tau}\right)^2 + \left(\frac{\partial Y}{\partial \tau}\right)^2 - \left(\frac{\partial X}{\partial \sigma}\right)^2 - \left(\frac{\partial Y}{\partial \sigma}\right)^2\right]. \tag{2.31}$$

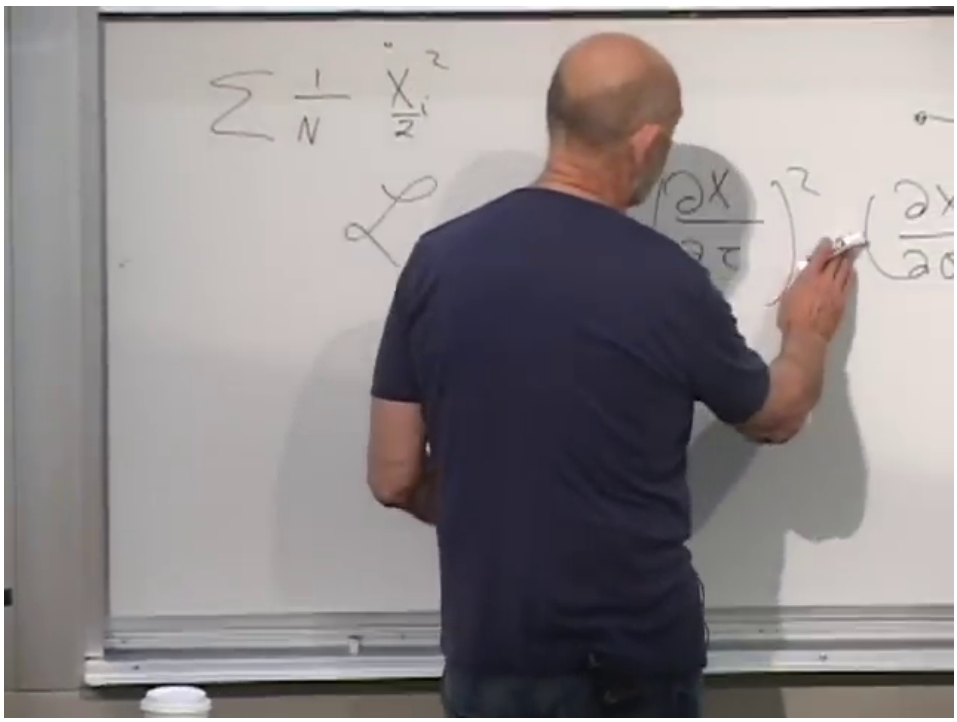

Figure 2.4: Continuum string Lagrangian. The board records the moment at which the minus sign of the potential term is being written.

The lecture pauses here to make an illuminating comparison. If we rename σ as a spatial coordinate on a line and rename X as a field ϕ, then this is just the energy of an ordinary wave system. Waves move left and right along the interval. When they hit the boundary, they reflect.

The boundary condition determines the character of the reflection. Under Dirichlet conditions the reflected wave flips sign. Under Neumann conditions it reflects without such inversion. Susskind says one could now write down the wave equation by varying the Lagrangian, but refuses to do so because the lecture will not need it. That refusal is part of the rhythm of the presentation, and we keep it. The wave viewpoint is useful; the full partial differential equation is not yet needed.

One more spoken clarification belongs here. The time variable τ is the rescaled time appropriate to the boosted description. Susskind loosely calls it proper time: the time carried along by the boosted clock so that we can continue to watch the internal motion instead of letting it freeze under time dilation.

Finally, the lecture reminds us why this energy matters. In the light-front description, the internal energy of the string will be identified, up to the same overall normalization convention as before, with the square of the relativistic mass of the string state.

2.6 Endpoint Dynamics and the Boundary-Condition Puzzle

Now the lecture returns to a question it deliberately postponed earlier. We have seen both Dirichlet and Neumann conditions. We have heard that free string endpoints should obey Neumann conditions. Why?

Here the lecture is at its cleanest. It raises the obstacle plainly and resolves it with a short Newton-law argument.

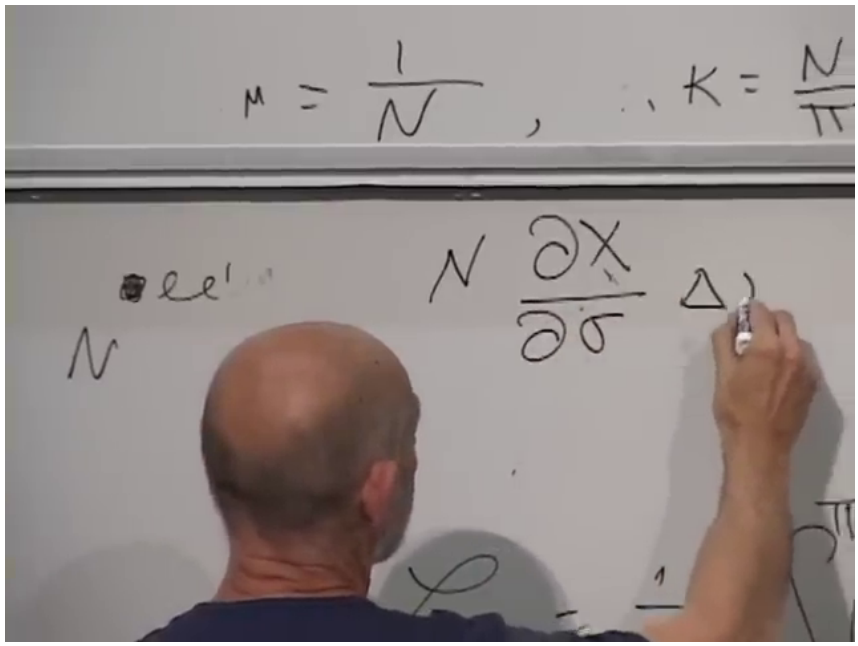

Figure 2.5: Discrete difference becomes derivative times spacing. This is the local step that converts the endpoint force into a boundary condition.

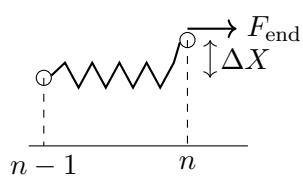

Figure 2.6: Endpoint geometry for the Newton-law argument. The last mass point feels a force from only one spring.

2.6.1 Question & Answer

Question. Why must the open-string endpoint satisfy Neumann rather than Dirichlet boundary conditions?

Answer. Because the endpoint is free rather than pinned, and for a free endpoint Newton's law forces the local stretch to vanish in the continuum limit.

Let the last bead be labeled n, and the previous one $n-1$. An interior bead is pulled from both sides. The endpoint is special: it feels only the force from the spring connecting it to the rest of the string. By Hooke's law,

$$F_{\text{end}} \sim k\,\Delta X. \tag{2.32}$$

Now we use the continuum replacement that the board is capturing:

$$\Delta X = \frac{\partial X}{\partial \sigma}\,\Delta\sigma. \tag{2.33}$$

Since $k \sim N$ and $\Delta\sigma \sim 1/N$, the N-dependence cancels, so

$$F_{\text{end}} \propto \frac{\partial X}{\partial \sigma}. \tag{2.34}$$

That is already the key step: the endpoint force is proportional to the slope of the string at the endpoint.

But Newton's law also says

$$F_{\text{end}} = \mu\,\frac{\partial^2 X}{\partial \tau^2}, \qquad \mu \sim \frac{1}{N}. \tag{2.35}$$

So if $\partial_\sigma X$ were nonzero at the endpoint, then the acceleration $\partial_\tau^2 X$ would grow like N as the continuum limit is taken. The endpoint would acquire an infinite acceleration. That is exactly the pathology the lecture wants to exclude.

The cure is immediate:

$$\left.\frac{\partial X}{\partial \sigma}\right|_{\sigma=0,\pi} = 0. \tag{2.36}$$

And the same argument applies to Y. Thus the free open-string endpoint satisfies Neumann boundary conditions.

This is a good example of the lecture's style. A boundary condition that could have been postulated as a formal rule is instead derived from a local mechanical argument. The mathematics is light, but the motivation is precise.

2.7 Fourier Modes, Harmonic Oscillators, and the First Quantum Spectrum

Now the system is finally ready to quantize. The first step is not yet quantum mechanics; it is a re-description of the classical string in terms of its harmonics.

Because the open string obeys Neumann conditions, we expand in cosines:

$$X(\sigma,\tau) = \sum_{n=0}^{\infty} X_n(\tau)\cos n\sigma, \qquad Y(\sigma,\tau) = \sum_{n=0}^{\infty} Y_n(\tau)\cos n\sigma. \tag{2.37}$$

The dependence on σ is carried by the cosines. The dependence on time τ sits in the coefficients $X_n(\tau)$ and $Y_n(\tau)$. That is where the string wiggles.

Differentiating with respect to time gives

$$\dot{X}(\sigma,\tau) = \sum_{n=0}^{\infty} \dot{X}_n(\tau)\cos n\sigma, \tag{2.38}$$

while differentiating with respect to σ gives

$$\frac{\partial X}{\partial \sigma} = -\sum_{n=1}^{\infty} n\, X_n(\tau) \sin n\sigma. \tag{2.39}$$

The board image suggests a sum beginning at $n = 0$, but the cleaned expression must begin at $n = 1$, since the differentiated zero mode vanishes.

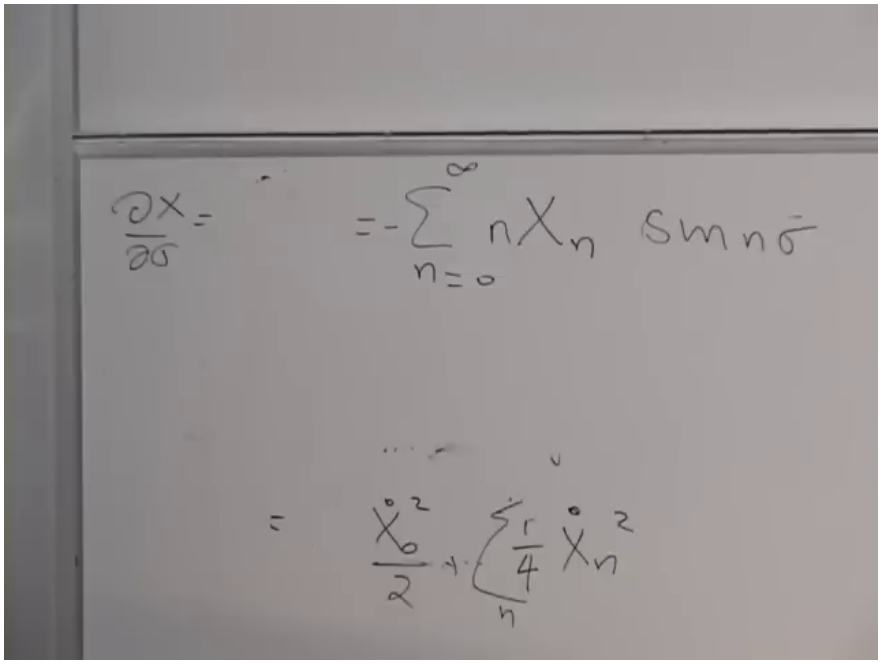

Figure 2.7: Mode expansion of the σ-derivative. The visible board evidence is the minus sign, the factor of n, and the passage from cosines to sines.

Now we insert the mode expansions into the energy. This is where the orthogonality formulas from the beginning of the lecture are harvested. All cross-terms with $n \neq m$ vanish after integration over σ. The double sums collapse to single sums. The $n = 0$ mode must be handled separately because its cosine norm is π rather than $\pi/2$.

For the kinetic piece of the X-sector we find

$$\frac{1}{2\pi}\int_0^{\pi} d\sigma\, \dot{X}^2 = \frac{1}{2}\dot{X}_0^2 + \frac{1}{4}\sum_{n=1}^{\infty} \dot{X}_n^2, \tag{2.40}$$

and for the stretching piece,

$$\frac{1}{2\pi}\int_0^{\pi} d\sigma \left(\frac{\partial X}{\partial \sigma}\right)^2 = \frac{1}{4}\sum_{n=1}^{\infty} n^2 X_n^2. \tag{2.41}$$

So the full X-sector energy is

$$E_X = \frac{1}{2}\dot{X}_0^2 + \frac{1}{4}\sum_{n=1}^{\infty}\left(\dot{X}_n^2 + n^2 X_n^2\right), \tag{2.42}$$

and the corresponding X-sector Lagrangian is

$$L_X = \frac{1}{2}\dot{X}_0^2 + \frac{1}{4}\sum_{n=1}^{\infty}\left(\dot{X}_n^2 - n^2 X_n^2\right). \tag{2.43}$$

The Y-sector is identical:

$$L_Y = \frac{1}{2}\dot{Y}_0^2 + \frac{1}{4}\sum_{n=1}^{\infty}\left(\dot{Y}_n^2 - n^2 Y_n^2\right). \tag{2.44}$$

Several things are now visible at once.

First, X_0 and Y_0 are the average positions of the string, hence the center-of-mass coordinates. They have kinetic energy but no restoring force. The zero mode is a free particle mode, not an oscillator.

Second, each mode with $n \geq 1$ is an ordinary harmonic oscillator. Comparing with the earlier oscillator formula, we read off

$$\omega_n = n, \qquad n \geq 1. \tag{2.45}$$

So the open string is a countable infinity of independent oscillators in the X-direction, together with another countable infinity in the Y-direction. These are nothing but the harmonics of the string.

Third, the internal vibrational energy is

$$E_{\text{int}} = \frac{1}{4}\sum_{n=1}^{\infty}\left(\dot{X}_n^2 + n^2 X_n^2 + \dot{Y}_n^2 + n^2 Y_n^2\right). \tag{2.46}$$

Up to the same light-front normalization convention introduced earlier, this is the quantity that becomes the relativistic mass squared of the string state. That is the bridge from oscillator mathematics back to particle physics.

Quantization is now easy. The lecture does not yet introduce creation and annihilation operators systematically; it merely uses the oscillator spectrum. For the n-th oscillator, if Q_n quanta are present and $\hbar = 1$, then the excitation energy above the unspecified ground-state offset is

$$\Delta E_n = Q_n\,\omega_n = Q_n\,n. \tag{2.47}$$

Susskind is careful to rename the occupation number Q so that it is not confused with the mode label n.

The lowest levels can then be counted directly.

Ground state. All oscillators are in their lowest state. This is not empty space. It is one open string with no internal excitation.

First excited level. Excite the $n = 1$ oscillator once. One may do this in the X-tower or in the Y-tower, so the first excited level has multiplicity 2.

Second excited level. Now the total excitation energy is 2. There are five possibilities:

- two quanta in the $n = 1$ X-oscillator,
- two quanta in the $n = 1$ Y-oscillator,

- one quantum in each of the $n = 1$ X- and Y-oscillators,
- one quantum in the $n = 2$ X-oscillator,
- one quantum in the $n = 2$ Y-oscillator.

So the second excited level has multiplicity 5.

This is exactly the point the lecture wanted to set up from the beginning. The open string has a discrete spectrum with appreciable gaps. It therefore behaves like a particle in the precise spectral sense explained earlier.

The final remarks of the lecture are a preview, not a full development. Open strings have endpoints, and once interactions are allowed, endpoints can join and split. A fluctuating open string may bring one of its endpoints into contact with another endpoint and thereby produce a closed string. So one may consistently have a theory of closed strings without open strings, but not a theory of open strings without closed strings. As a rough guide only, the lecture says that open strings often behave like photons, while closed strings behave like gravitons.

2.8 Summary

The lecture unfolds by preparation, obstacle, and payoff. It first stocks the blackboard with the mathematics of functions on $[0, \pi]$: discretization, boundary conditions, Fourier expansions, orthogonality, and the harmonic oscillator. It then sharpens the notion of a particle by replacing pointlikeness with spectral isolation. With that conceptual and mathematical groundwork in place, the infinite-momentum or light-front viewpoint recasts the relativistic string as an exact two-dimensional mechanical system. The open string is built as a chain of masses and springs, passed to a continuum description, fixed by a Neumann endpoint condition derived from Newton's law, and finally rewritten as an infinite collection of uncoupled harmonic oscillators. The resulting spectrum is discrete, gapped, and therefore particle-like. That is the central gain of the lecture, and it is the point from which the next step, the closed string, naturally begins.

Chapter 3

String Theory and M-Theory

Susskind begins this lecture in the way he says he has come to like: not by attacking the main problem head on, but by clearing the ground with two preliminaries. First comes a rapid harmonic-oscillator review, done not because the audience has never seen it, but because the notation for string modes must be fixed before anything else can be interpreted. Then comes a review of the spin of massless particles. Only after those two pieces are in place does the lecture turn to the low-lying spectrum of the string. We keep that order here. The transcript is the primary guide, while the preserved blackboard frames serve as local evidence for equations and board layout. In that spirit these notes explicitly preserve the lecture of Leonard Susskind and the supporting curation work of LazyingArt LLC and Video2Book.

3.1 Rapid Harmonic-Oscillator Review for String Modes

We begin exactly where the lecture begins: with the warning that this is only a quick reset of notation. The nth mode of the open string behaves like a harmonic oscillator, and the index n is the same n that appears in the mode function $\cos n\sigma$. It counts how many oscillations fit along the interval $0 \leq \sigma \leq \pi$. The only unusual feature is a normalization inherited from the string calculation: a factor of 4 appears in the denominator, and Susskind is quite explicit that we should simply keep it.

The first blackboard frame is useful not because it reveals every symbol perfectly, but because it preserves the comparison the lecture is making. The upper and lower lines contain the same quadratic pieces, and only the sign of the potential term changes. The right-hand labels are cropped, so we stabilize them from the transcript:

$$E_n = \frac{\dot{X}_n^2}{4} + \frac{n^2 X_n^2}{4}, \tag{3.1}$$

$$L_n = \frac{\dot{X}_n^2}{4} - \frac{n^2 X_n^2}{4}. \tag{3.2}$$

The point is simple: the nth mode is an oscillator of frequency n. That is what we want to carry forward.

Now we do the first standard step toward quantization. At the board Susskind suppresses the label

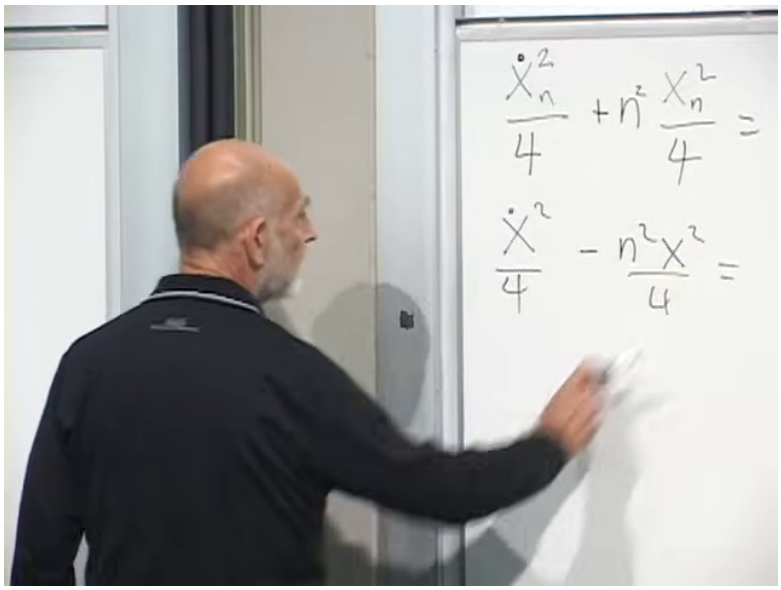

Figure 3.1: Harmonic oscillator energy and Lagrangian.

n for a moment and writes the single oscillator coordinate simply as x. From the Lagrangian,

$$p = \frac{\partial L}{\partial \dot{x}} = \frac{\dot{x}}{2}, \tag{3.3}$$

so $\dot{x} = 2p$. Substituting back into the energy gives the Hamiltonian

$$H = p^2 + \frac{n^2 x^2}{4}. \tag{3.4}$$

So far nothing mysterious has happened. We have merely rewritten the same oscillator in Hamiltonian form.

3.2 Hamiltonian, Ladder Operators, and the Return to the String

The next move is the usual oscillator trick, and Susskind presents it in the usual brisk, heuristic way. Since

$$H = p^2 + \left(\frac{nx}{2}\right)^2, \tag{3.5}$$

we are naturally led to the complex combinations

$$\frac{nx}{2} \pm ip. \tag{3.6}$$

At the blackboard this is phrased as a factorization of the sum of squares,

$$H \sim \left(\frac{nx}{2} + ip\right)\left(\frac{nx}{2} - ip\right), \tag{3.7}$$

and the point is not a polished discussion of operator ordering. The point is that these are the combinations out of which the creation and annihilation operators are built.

To normalize them we must check the commutator. With $\hbar = 1$,

$$[x, p] = i. \tag{3.8}$$

Let us check what the tentative pair does:

$$\left[\frac{nx}{2} + ip, \frac{nx}{2} - ip\right] = -\frac{ni}{2}[x, p] + \frac{ni}{2}[p, x] \tag{3.9}$$

$$= -\frac{ni}{2}(i) + \frac{ni}{2}(-i) \tag{3.10}$$

$$= \frac{n}{2} + \frac{n}{2} = n. \tag{3.11}$$

So the natural combinations almost work, but not quite. Their commutator is n, not 1. The fix is immediate: divide by $\sqrt{n}$. Thus

$$a^- = \frac{1}{\sqrt{n}}\left(\frac{nx}{2} + ip\right) = \frac{\sqrt{n}\,x}{2} + \frac{i}{\sqrt{n}}p, \tag{3.12}$$

$$a^+ = \frac{1}{\sqrt{n}}\left(\frac{nx}{2} - ip\right) = \frac{\sqrt{n}\,x}{2} - \frac{i}{\sqrt{n}}p, \tag{3.13}$$

and now

$$[a^-, a^+] = 1. \tag{3.14}$$

The lecture makes an instructive point here. One cannot read off from the visible minus sign alone which operator raises the energy and which lowers it. The criterion is the commutator. The convention above is the one for which a^+ is the raising operator and a^- is the lowering operator.

Once the string is taken to move with very large momentum along the z-axis, the independent oscillations are transverse. In three space dimensions there are therefore two copies of the oscillator, one in the x-direction and one in the y-direction. We introduce a second pair,

$$b^- = \frac{\sqrt{n}\,y}{2} + \frac{i}{\sqrt{n}}p_y, \tag{3.15}$$

$$b^+ = \frac{\sqrt{n}\,y}{2} - \frac{i}{\sqrt{n}}p_y, \tag{3.16}$$

with

$$[b^-, b^+] = 1. \tag{3.17}$$

The second blackboard frame is especially valuable because it shows the lecture's organization: first the commutator block, then a dividing line, and only then the explicit operator formulas. In cleaned form the upper block is

$$[a^-, a^+] = 1, \qquad [b^-, b^+] = 1, \qquad [x, p] = i. \tag{3.18}$$

The lower part of the frame only begins the formula for a^-, so the full coefficients are stabilized from the transcript.

Now comes the step that turns the oscillator algebra back into the string. Adding the two defining equations for $a_n^\pm$ eliminates the momentum term and gives

$$a_n^+ + a_n^- = \sqrt{n}\,x_n. \tag{3.19}$$

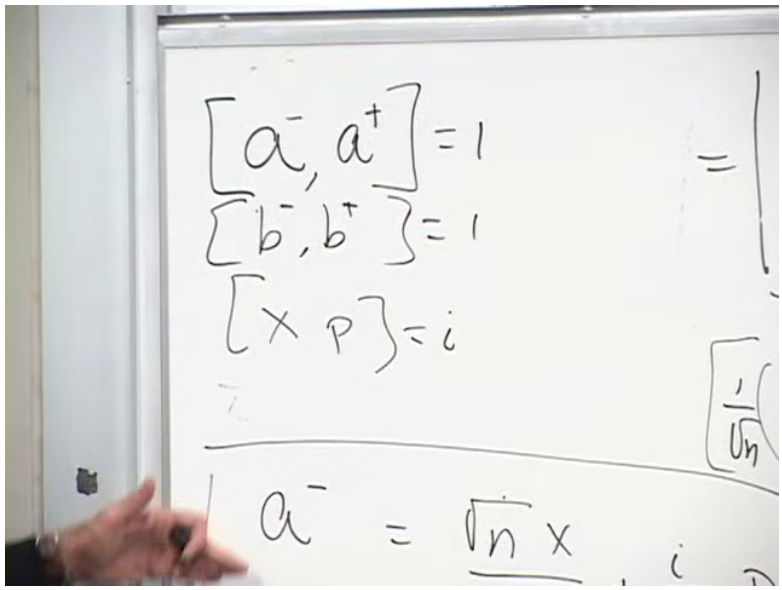

Figure 3.2: Commutators for the a- and b-oscillator modes.

Hence

$$x_n = \frac{a_n^+ + a_n^-}{\sqrt{n}}, \qquad y_n = \frac{b_n^+ + b_n^-}{\sqrt{n}}. \tag{3.20}$$

This is not bookkeeping for its own sake. It is the bridge back from the abstract oscillator to the string coordinate itself.

The open string is parameterized by $\sigma \in [0, \pi]$. Its transverse coordinates are expanded in cosine modes,

$$x(\sigma, \tau) = \sum_n x_n(\tau) \cos n\sigma, \tag{3.21}$$

$$y(\sigma, \tau) = \sum_n y_n(\tau) \cos n\sigma. \tag{3.22}$$

Why cosines and not sines? Because the endpoint condition is

$$\partial_\sigma x\big|_{\sigma=0,\pi} = 0, \qquad \partial_\sigma y\big|_{\sigma=0,\pi} = 0, \tag{3.23}$$

so the slope vanishes at the endpoints. Substituting the operator expressions for the Fourier coefficients yields

$$x(\sigma, \tau) = \sum_n \frac{a_n^+ + a_n^-}{\sqrt{n}} \cos n\sigma, \tag{3.24}$$

$$y(\sigma, \tau) = \sum_n \frac{b_n^+ + b_n^-}{\sqrt{n}} \cos n\sigma. \tag{3.25}$$

This is why the lecture spent time on the oscillator review. Once the coordinate operators are written in terms of creation and annihilation operators, the action of those coordinates on a quantum state becomes transparent.

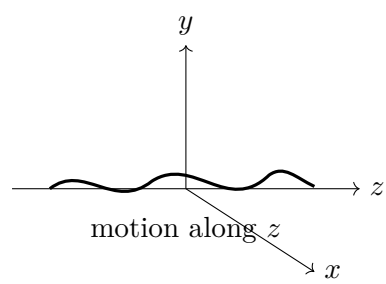

Figure 3.3: The string is taken to move with large momentum along the z-axis. The independent oscillations are transverse, in the x- and y-directions.

3.2.1 Question & Answer

Question. Do the coefficients x_n depend on σ?

Answer. No. All of the σ-dependence is carried by the cosine functions. The coefficients x_n and y_n are Fourier coefficients. In the present discussion they depend on the time variable τ, because the modes themselves oscillate in time. The dots in the oscillator formulas are therefore τ-derivatives, not derivatives with respect to the spatial parameter σ.

3.3 Massive Versus Massless Spin

With the notation now under control, the lecture turns to the second preliminary. The question is not merely how many states a spin-J system has, but why the answer changes when the particle is massless.

For a massive particle,

$$N_{\text{massive}}(J) = 2J + 1. \tag{3.26}$$

Thus spin 0 has one state, spin 1 has three, spin 2 has five, and so forth. But for massless particles the pattern changes sharply:

$$N_{\text{massless}}(0) = 1, \qquad N_{\text{massless}}(s > 0) = 2. \tag{3.27}$$

Spin 1 no longer has three states; it has two. Spin 2 no longer has five states; it also has two. Spin 0 remains the special case.

The way Susskind builds this is through motion along a chosen axis, say the z-axis. A spin-1 particle can have spin component $+1$, 0, or -1 along that axis. A spin-2 particle can have $+2, +1, 0, -1, -2$. Reflection symmetry already guarantees that maximal positive and maximal negative helicity must come together. The real issue is whether the intermediate states are forced on us.

For a massive particle they are forced on us, because we can go to its rest frame. One imagines following the particle until it is at rest. Once it is at rest, rotation invariance tells us that if one spin orientation is allowed, then rotated ones are allowed as well. We can then boost back to a moving frame and obtain states whose spin is no longer simply maximal or minimal along the original direction of motion. That is why the full multiplet must be present for a massive particle.

For a massless particle the argument collapses at the first step. There is no rest frame. One cannot bring the particle to rest, rotate it there, and then boost back. The missing rest frame is exactly what allows the intermediate states to disappear. In the massless case it is consistent to keep only the two helicity states of maximal and minimal spin along the direction of motion.

3.3.1 Question & Answer

Question. Why do massless particles have only two polarization states?

Answer. Because the rotate-then-boost argument requires a rest frame. A massive particle can be brought to rest, and once it is at rest the spin can be reoriented and then boosted back into motion, generating the full $2J + 1$ multiplet. A massless particle cannot be brought to rest, so that construction is unavailable. The theory can therefore consistently retain only the two extreme helicity states.

3.4 Photon Polarization as the Model Case

The lecture now translates the abstract spin discussion into the familiar language of photons. This is not a change of topic. It is the concrete model for what will happen to the first excited open-string states a few minutes later.

A photon moving along the z-axis is polarized in the transverse xy-plane. The electric field never points along the direction of motion; Maxwell's equations forbid such a longitudinal polarization. Thus the photon has two transverse basis states, which we may call $|x\rangle$ and $|y\rangle$. These are linearly polarized states. From them we build the circular basis,

$$|R\rangle = |x\rangle + i|y\rangle, \qquad |L\rangle = |x\rangle - i|y\rangle. \tag{3.28}$$

These are not normalized states here; they are the lecture's operational basis vectors, and that is enough for the present purpose.

What matters is that the circular combinations are helicity eigenstates. They are the two genuine spin states of the photon. Linear polarization is different. A state such as $|x\rangle + |y\rangle$ is still only a superposition of the same two physical states; it is simply linearly polarized at a rotated angle in the transverse plane. The complex unit i is what converts the linear basis into the circular one.

The same logic also gives the useful graviton analogy. A massless spin-2 particle again has only two states. One may think of them, heuristically, as the analog of two photons of like helicity combined. That way of speaking is useful for spin counting, but the lecture is careful not to turn it into a literal claim that a graviton is made of two photons.

3.4.1 Question & Answer

Question. Does linear polarization mean zero angular momentum?

Answer. Not as a sharp quantum number. A linearly polarized photon is a superposition of the two helicity eigenstates. Its average angular momentum along the direction of motion may vanish,

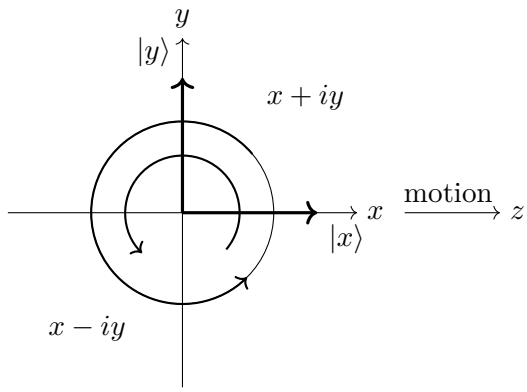

Figure 3.4: Linear and circular polarization in the plane transverse to the motion.

but a measurement never produces a third state of zero helicity. It produces one helicity or the other.

3.5 The First Excited Open-String States and the Photon-Like Interpretation

At this point the lecture cashes in both preliminaries. Susskind explicitly recaps: now we know something about harmonic oscillators, and now we know something about spin. So now we can look at the spectrum.

Before listing states, he reminds us of one essential point of interpretation. In this formal analogy between the string oscillators and relativistic particle states, the oscillator energy contributes to m^2, not directly to m. So when we say that an excitation adds one unit of energy, we mean one unit of mass-squared in the lecture's units.

Let $|0\rangle$ denote the unexcited string state. This is not empty space. It is a single string whose oscillators are all as quiet as quantum mechanics allows:

$$a_n^-|0\rangle = 0, \qquad b_n^-|0\rangle = 0 \tag{3.29}$$

for every mode n. The lecture deliberately leaves its mass-squared undetermined:

$$m^2(|0\rangle) = m_0^2. \tag{3.30}$$

That unknown is the stage on which the next argument plays out.

What is the cheapest thing we can do to the string? We apply a raising operator to the oscillator of smallest frequency, namely the $n = 1$ mode. This gives two states,

$$a_1^+|0\rangle, \qquad b_1^+|0\rangle. \tag{3.31}$$

Since the nth oscillator adds n units, these states have

$$m^2 = m_0^2 + 1. \tag{3.32}$$

The lecture briefly pauses here to note that the actual physical scale hidden in that "1" has not yet been discussed. The units are missing and will be taken up later. But the internal logic does not wait for that discussion.

There is another essential point. The states $a_1^+|0\rangle$ and $b_1^+|0\rangle$ are degenerate, so any linear combination of them is again an eigenstate with the same value of m^2. In particular,

$$\left(a_1^+ \pm i\, b_1^+\right)|0\rangle \tag{3.33}$$

gives the circularly polarized combinations.

Now the transformation argument arrives. These operators came from the transverse coordinates x and y. Under rotations about the preferred z-axis they mix exactly as the components of a two-dimensional vector mix. So the first excited open-string states transform exactly like the two transverse polarization states of a photon. There is no third state of the same energy, because there is no third independent transverse oscillator in this three-dimensional discussion.

That is the crucial payoff. If an object carries only two transverse polarization states, then the only relativistically consistent interpretation is that it is massless. A massive spin-1 particle would have to carry a third polarization state, because it could be brought to rest and rotated. Therefore these states must be massless. The conclusion is

$$m_0^2 + 1 = 0. \tag{3.34}$$

So the first excited open-string states are photon-like, or more generally gauge-boson-like, and they are necessarily massless.

The lecture is careful with the language here. It is too fast, and too model-dependent, to say that we have found *the* photon. What we have found is a photon-like object. Historically this mattered. In the old hadronic interpretation of string theory, such a massless spin-1 state was viewed as a wart rather than a feature, because there are no massless mesons. But in the present logical arc it is a success first. Only then does it become a problem.

3.6 The Tachyonic Ground-State Disaster

And now the sharp turn. The first excited state has come out beautifully: it is massless, transverse, and photon-like. But the very condition that makes it so,

$$m_0^2 + 1 = 0, \tag{3.35}$$

forces the ground state to have

$$m_0^2 = -1 \tag{3.36}$$

in the lecture's units. So the success of the first excited state immediately produces the failure of the ground state. This is the moment in the lecture where Susskind says, with deliberate drama, that perhaps we should simply throw the theory away.

3.6.1 Question & Answer

Question. Does negative mass-squared mean faster-than-light propagation?

Answer. Not in the sense relevant to real physics. The name "tachyon" arose from a naive reading of the dispersion relation, but the lesson the lecture wants us to keep is different. Negative mass-squared signals an instability of the vacuum. The disease is not useful superluminal signaling; it is that we have expanded the theory around the wrong configuration.

To see how the older language arose, Susskind first recalls the group-velocity rule

$$v_g = \frac{dE}{dp}. \tag{3.37}$$

For a nonrelativistic particle,

$$E(p) = \frac{p^2}{2m}, \qquad \frac{dE}{dp} = \frac{p}{m}, \tag{3.38}$$

which is just the ordinary velocity. For a relativistic particle, in units with $c = 1$,

$$E = \sqrt{p^2 + m^2}, \qquad \frac{dE}{dp} = \frac{p}{\sqrt{p^2 + m^2}}. \tag{3.39}$$

As long as $m^2 > 0$, this is less than 1. If one naively flips the sign of m^2, the algebra suggests a velocity larger than 1, and that is why the old term "tachyon" stuck.

But now the lecture changes the question. What does negative mass-squared really mean? The answer is not in the kinematics of a single particle, but in the stability of the field configuration. The safe relation to extract from the somewhat sign-corrected wave-equation discussion is

$$\omega^2 = k^2 + m^2. \tag{3.40}$$

Correspondingly, the quadratic part of the field energy is

$$\mathcal{E}_\phi = \frac{1}{2}\left[(\partial_t\phi)^2 + (\partial_x\phi)^2 + m^2\phi^2\right]. \tag{3.41}$$

If $m^2 > 0$, then displacing the field costs positive potential energy, and the system behaves like an ordinary harmonic oscillator. If $m^2 < 0$, the quadratic potential near $\phi = 0$ is inverted:

$$V(\phi) \propto -|m|^2\phi^2. \tag{3.42}$$

So $\phi = 0$ is not a minimum at all. It is the top of a hill.

This is why the lecture turns to the pencil, pendulum, and domino analogies. A perfectly balanced pencil can in principle sit forever, but the slightest push sends it away. A row of coupled inverted pendulums behaves similarly: disturb one of them, and an instability front propagates through the system. Yet that front does not outrun the characteristic wave speed. The instability spreads causally. So the right reading of the tachyon is not "faster than light," but "unstable vacuum."

The lecture then makes a very pointed comparison. A quantum theory with an upside-down potential is not nonsense in the formal sense. One can write it down. What it lacks is a ground state. That is the real problem. The bosonic open string, taken in this simple form, does not look like the world. It gives us a beautiful photon-like state, but it gives it to us on top of an unstable vacuum. That is why one eventually needs superstring theory.

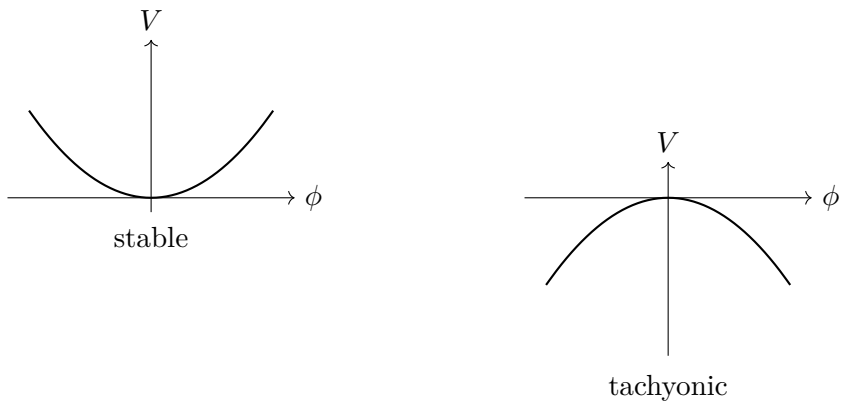

Figure 3.5: Positive mass-squared gives an ordinary quadratic potential; negative mass-squared turns the quadratic term upside down and makes $\phi = 0$ unstable.

3.7 Rotating Strings, Joining and Splitting, Closed Strings, and Gravity

Before the final pivot to gravity, the lecture pauses to note that the open-string state we have just identified is not merely a formal polarization count. Once one introduces charged string states and the interaction rules, the emission and absorption patterns match the familiar behavior of gauge bosons. That thread is not pursued mathematically here, because Susskind wants to move to the next indispensable idea: closed strings.

He begins that move by reminding us that the free string can do more than vibrate in simple transverse modes. It can also rotate, and when it rotates it carries angular momentum. In the quantum theory each extra unit of angular momentum costs extra energy. So the string naturally supports towers of higher-spin excitations. This matters because it prepares us to think of string states not just as particles, but as particles with very definite transformation properties.

Now comes the interaction rule, presented quite explicitly as a qualitative one. We are not yet doing full mathematics with it. The basic open-string process is this: whenever two endpoints meet in space, there is an amplitude for them to join and form a single open string. By reversibility, there must also be an amplitude for the inverse process, in which an open string breaks and produces two endpoints. The parameter controlling the size of these amplitudes is the string coupling constant.

The lecture also makes a practical point about weak coupling. At small coupling, strings are effectively transparent to one another. They may pass through each other without doing anything. If they happen to touch, there is a small probability per unit time for joining to occur. That probability is, at this level, the same basic one for all string states. Differences in observed joining rates come from how long the strings spend in contact.

Now the decisive topological point arrives. An open string can curve around and encounter its own tail. If the joining process is allowed at all, then nothing prevents the string from joining to itself and forming a closed loop. So the existence of interacting open strings already implies the existence of closed strings. Closed strings are not optional. They are forced on us.

What about the opposite direction? Must a theory of closed strings also contain open strings? The lecture is clear: no. Closed strings have their own interaction channel. Two closed strings may

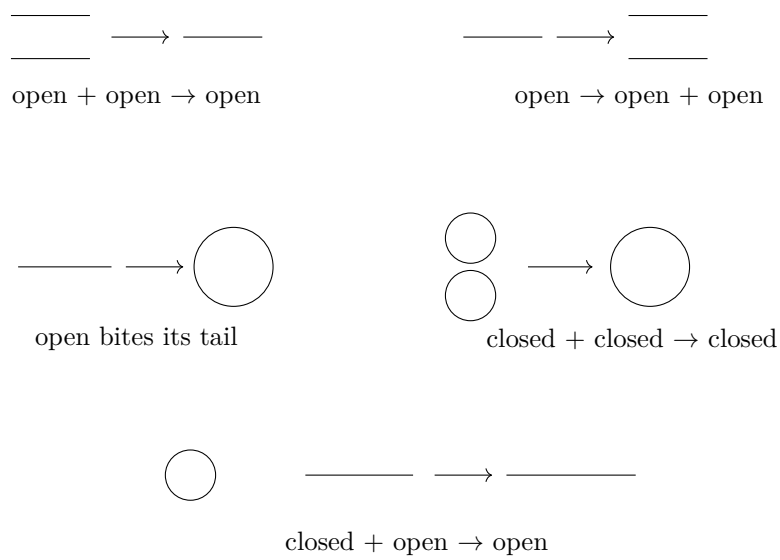

Figure 3.6: Qualitative interaction rules used in the lecture: joining, splitting, self-closing, pure closed-string interaction, and absorption of a closed string by an open string.

meet, interact, and continue as closed strings without ever producing open endpoints. So a pure closed-string theory is possible, but a pure interacting open-string theory is not.

The lecture then sharpens the point further. If the lowest interesting excitation of the open string is photon-like, what should we expect from the lowest interesting excitation of the closed string? The answer is the graviton. This is not derived in detail here, but Susskind makes the structural claim very strongly: every string theory necessarily contains closed strings, and therefore every string theory necessarily contains gravity.

There is still one more physical payoff. A closed string can interact with an open string and be absorbed, leaving behind a larger open string. Interpreted in particle language, that is the absorption of a graviton by a non-graviton. Likewise a closed string can interact with another closed string. In this simple and very physical sense, everything can absorb a graviton. That is one of the lecture's ways of expressing the universality of gravity.

3.7.1 Question & Answer

Question. Why must interacting open-string theories contain closed strings, while pure closed-string theories need not contain open strings?

Answer. Because the very joining rule that lets open strings interact also lets a single open string join to itself and form a closed loop. Once open strings interact, closed strings are automatically generated. Closed strings, however, already possess a direct closed-string interaction channel, so they can interact among themselves without producing open endpoints. Thus pure closed-string theories are possible, but interacting open-string theories cannot remain purely open.

3.8 Summary

The lecture is a deliberately staged argument. We first reset the notation by treating each string mode as a harmonic oscillator with the lecture's characteristic factors of 1/4. We then convert those oscillators into ladder operators and finally back into the string coordinates themselves. Next we review the special spin structure of massless particles and make the photon the model example. Only then do we return to the open-string spectrum. The first excited open-string states transform as a transverse vector with exactly two polarizations, so they must be massless and photon-like. That apparent success immediately forces the ground state to have negative mass-squared, and the lecture turns that crisis into the correct physical lesson: a tachyon means an unstable vacuum, not a practical violation of relativity. Finally, to reach gravitons we move from free open strings to interacting strings, discover that interacting open strings inevitably generate closed strings, and thereby discover that gravity is not an optional extra in string theory. The question of units is left where the lecture leaves it: postponed, but now unavoidable.

Chapter 4

String Theory and M-Theory

We want to move on to closed strings, but the lecture quite deliberately does not begin with the loop itself. First comes a small reminder from canonical mechanics, because a symmetry argument will later be needed and it is better to have that tool already on the board. Only then do we draw the closed string, mark a point $\sigma = 0$, and ask the question that governs the whole chapter: is that marked point a physical feature, or only a coordinate convention? From that question come the doubled left- and right-moving oscillators, the level-matching rule, and the appearance of a massless spin-two particle.

4.1 Noether's Theorem as a Prelude

Before coming to closed strings, Susskind pauses for what he calls a little bit of mathematics. The pause matters. He is not changing the subject; he is loading the machinery that will later explain a constraint on the closed-string spectrum.

Let the Lagrangian depend on coordinates q_i and their velocities $\dot{q}_i$. Then the canonical momentum conjugate to q_i is

$$p_i = \frac{\partial L}{\partial \dot{q}_i}. \tag{4.1}$$

Now take an infinitesimal transformation

$$q_i \longrightarrow q_i + \delta q_i. \tag{4.2}$$

Noether's theorem associates to this transformation a conserved quantity, or in quantum mechanics a generator. In the schematic form needed here,

$$Q_{\text{Noether}} = \sum_i p_i \, \delta q_i. \tag{4.3}$$

That is the formula he wants on the board.

As a familiar example, one may think of a small rotation in the xy-plane,

$$\delta x = y \, \epsilon, \qquad \delta y = -x \, \epsilon, \tag{4.4}$$

with ϵ a small angle. The corresponding Noether charge is angular momentum. We do not need the example for its own sake; we only need the habit of mind. A symmetry acts by shifting coordinates, and the quantity built from momentum times displacement is the generator of that shift.

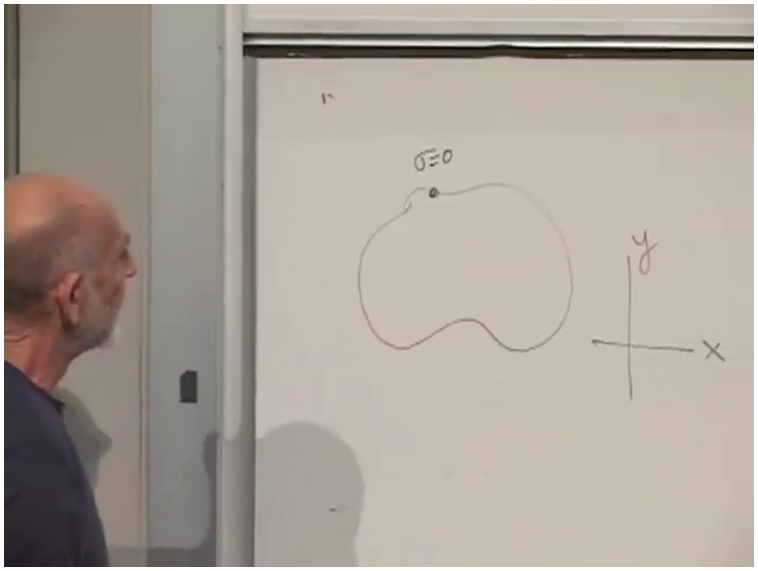

Figure 4.1: Closed string with a marked $\sigma = 0$ point.

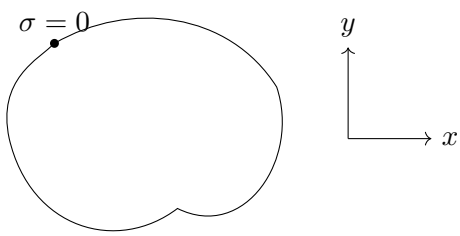

Figure 4.2: Clean schematic of the same setup.

That is all the preparation we need. Susskind now does exactly what he says he will do: let us forget the mathematics for the moment and come to closed strings.

4.2 Closed Strings, Marked Points, and Intrinsic Orientation

A closed string is simply a string without ends. For an open string we used a coordinate σ running from 0 to π. For a closed string the natural range is

$$0 \leq \sigma \leq 2\pi. \tag{4.5}$$

One point on the loop is chosen and labeled $\sigma = 0$. Halfway around lies $\sigma = \pi$, and so on. But the lecture immediately insists on the obvious and important point: which point we call $\sigma = 0$ really does not matter.

The blackboard picture is only a loop in the xy-plane with one marked point. Nothing in the shape distinguishes that point geometrically. The mark is a bookkeeping device, not a physical scar on the string. A clean redraw makes the same point with less blackboard noise:

Now comes the first conceptual caution of the lecture. The orientation of the string is *not* its clockwise or anticlockwise orientation in the plane. We could lift the rubber band off the blackboard,

turn it over, and put it back; the spatial clockwise sense would reverse, but the intrinsic direction of increasing σ along the string would still be perfectly meaningful. When Susskind speaks of orientation, he means this intrinsic directionality along the loop.

That is why, when he next speaks of right-moving and left-moving waves, "right" and "left" do not refer to directions in the room. A right-moving disturbance is one that moves toward larger σ; a left-moving disturbance moves toward smaller σ. On an open string such disturbances reflect from the ends and are not separately conserved. On a closed string they can run freely around and around the loop.

4.3 Periodic Boundary Conditions and Fourier Modes

With the geometry in place, we translate it into the usual coordinate language. The closed string is described by functions $x(\sigma)$ and $y(\sigma)$, just as the open string was described by transverse coordinates depending on σ. The only new ingredient is periodicity:

$$x(0) = x(2\pi), \qquad y(0) = y(2\pi). \tag{4.6}$$

These are the closed-string boundary conditions. There are no ends, but there is still a condition relating the values at $\sigma = 0$ and $\sigma = 2\pi$, because those are the same point.

A periodic function on the interval $0 \leq \sigma \leq 2\pi$ has the Fourier expansion

$$x(\sigma) = \sum_{n\in\mathbb{Z}} x_n e^{in\sigma}. \tag{4.7}$$

The fact that n must be an integer is nothing mysterious: $e^{in\sigma}$ returns to itself at $\sigma = 2\pi$ only when $n \in \mathbb{Z}$. Susskind then rewrites the expansion by separating the positive, negative, and zero modes:

$$x(\sigma) = x_0 + \sum_{n>0} x_n e^{in\sigma} + \sum_{n>0} x_{-n} e^{-in\sigma}, \tag{4.8}$$

$$y(\sigma) = y_0 + \sum_{n>0} y_n e^{in\sigma} + \sum_{n>0} y_{-n} e^{-in\sigma}. \tag{4.9}$$

The zero modes x_0 and y_0 are not oscillations at all. They give the center-of-mass position of the string. The other coefficients are the real dynamical degrees of freedom. Their time dependence is buried in the mode amplitudes $x_n(\tau)$ and $y_n(\tau)$.

At this stage the lecture wants us to think physically, not formally. Positive and negative mode numbers encode the two intrinsic directions of propagation along the string. A disturbance with $n > 0$ moves one way around the loop and a disturbance with $n < 0$ moves the other way. The sign of n is therefore not just algebraic decoration; it remembers the sense of motion along σ.

It is also useful to keep in mind that for a mode labeled by $\pm n$ the frequency is proportional to $n > 0$, exactly as it was for the open string. Higher $|n|$ means shorter wavelength and therefore higher frequency.

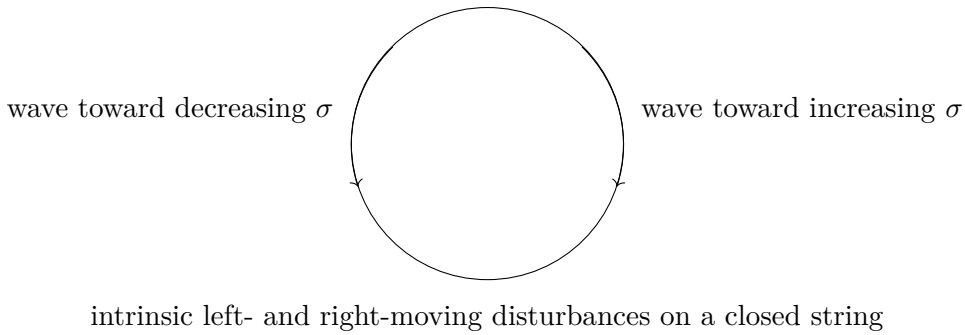

Figure 4.3: A cleaned companion picture of the two running-wave sectors.

4.4 Left- and Right-Moving Energies and the Doubled Oscillator Structure

Now we pass from the kinematics of periodic functions to the dynamics. The Lagrangian is the same one we used for the open string, except that the σ-integration now runs from 0 to 2π:

$$L \sim \int_0^{2\pi} d\sigma \Big[(\partial_\tau x)^2 - (\partial_\sigma x)^2 + (\partial_\tau y)^2 - (\partial_\sigma y)^2\Big]. \tag{4.10}$$

The corresponding energy is obtained, as Susskind says on the board, by changing the relative minus sign to a plus sign:

$$E \sim \int_0^{2\pi} d\sigma \Big[(\partial_\tau x)^2 + (\partial_\sigma x)^2 + (\partial_\tau y)^2 + (\partial_\sigma y)^2\Big]. \tag{4.11}$$

Then comes a very revealing rewrite. On the board Susskind remarks that he is off by a factor of two; in cleaned form the identity is

$$(\partial_\tau X)^2 + (\partial_\sigma X)^2 = \frac{1}{2}\Big[(\partial_\tau X + \partial_\sigma X)^2 + (\partial_\tau X - \partial_\sigma X)^2\Big], \tag{4.12}$$

and similarly for Y. This is not just algebra. It separates the energy into two pieces, one carried by one family of running waves and one carried by the other. Up to a choice of convention about which sign is called left and which is called right, a disturbance depending on $\sigma + \tau$ moves one way and a disturbance depending on $\sigma - \tau$ moves the other.

Once the Fourier expansion is plugged into the Lagrangian, each mode amplitude becomes a harmonic oscillator. The key difference from the open string is that each frequency now comes in two directional copies. The closed string therefore has twice as many oscillator families.

Here the notation becomes a little ugly, and Susskind says so himself. We keep his notation, but cleanly. In the x-sector the creation operators are

$$a_n^+, \qquad a_{-n}^+, \tag{4.13}$$

and in the y-sector they are

$$b_n^+, \qquad b_{-n}^+. \tag{4.14}$$

The superscript + means "creation operator." The sign of the subscript records the propagation direction. The symbol a versus b tells us whether the excitation is in the x-direction or the y-direction. This bookkeeping is clumsy, but it is the bookkeeping of the lecture, and it encodes the essential point: a closed string has a doubled oscillator structure because it supports both left-moving and right-moving running waves.

4.5 The First Spectrum Puzzle and the Level-Matching Rule

The lecture now pivots by recap. Before trying to understand the closed-string spectrum, Susskind reminds us what happened for the open string. There the lowest nontrivial excitations were created by

$$a_1^+|0\rangle, \qquad b_1^+|0\rangle, \tag{4.15}$$

and the circular combinations

$$(a_1^+ \pm i b_1^+)|0\rangle \tag{4.16}$$

carried angular momentum ± 1 about the direction of motion. That was the photon story.

If we imitate this too quickly for the closed string, the first states we would write down are

$$a_1^+|0\rangle, \qquad a_{-1}^+|0\rangle, \qquad b_1^+|0\rangle, \qquad b_{-1}^+|0\rangle. \tag{4.17}$$

At first glance this looks perfectly reasonable. It seems to say that we now have two copies of the old photon-like excitation, one associated with one direction of motion along the string and one associated with the other. Susskind even pushes the naïve picture far enough to note that these would have to behave like massless particles, because there is no angular-momentum-zero partner among them.

And then he abruptly says: this is not the right picture.

He makes the tension sharper by looking one level higher. If one simply keeps writing down all states of the next energy, one gets a mess: too many states, the wrong combinations, and no sensible organization into rotational multiplets. That is the point at which he tells us that not every oscillator expression we can write corresponds to a legitimate state of the closed string.

4.5.1 Question & Answer

Question. Why is the naïve doubled-photon picture not the right closed-string spectrum?

Answer. Because the closed string comes with a constraint. One does *not* keep every state that can be written down from the oscillators. The rule, given first as a postulate and only later derived, is the level-matching condition:

$$E_L = E_R. \tag{4.18}$$

The total left-moving energy must equal the total right-moving energy.

That single rule immediately kills the four one-oscillator states above. Each of them carries energy only in one running-wave sector. The same thing happens to the obvious energy-two states

$$a_2^+|0\rangle, \qquad a_{-2}^+|0\rangle, \qquad b_2^+|0\rangle, \qquad b_{-2}^+|0\rangle, \tag{4.19}$$

because these too carry two units of energy in only one direction. In the lecture Susskind quite literally crosses such states out. The rule is presented first as an unexplained but necessary restriction, and only later are we told why it is true.

state	angular momentum M
$(a_1^+ + ib_1^+)(a_{-1}^+ + ib_{-1}^+)\|0\rangle$	$+2$
$(a_1^+ - ib_1^+)(a_{-1}^+ - ib_{-1}^+)\|0\rangle$	-2
$(a_1^+ + ib_1^+)(a_{-1}^+ - ib_{-1}^+)\|0\rangle$	0
$(a_1^+ - ib_1^+)(a_{-1}^+ + ib_{-1}^+)\|0\rangle$	0

Table 4.1: Lowest level-matched closed-string states in the circular basis.

4.6 Massless Spin-2, Dilaton, and Axion

Once level matching is imposed, the first nontrivial surviving states appear at total excitation energy two:

$$a_1^+ a_{-1}^+|0\rangle, \qquad b_1^+ b_{-1}^+|0\rangle, \qquad a_1^+ b_{-1}^+|0\rangle, \qquad a_{-1}^+ b_1^+|0\rangle. \tag{4.20}$$

Each one carries one unit of right-moving energy and one unit of left-moving energy. These are therefore legitimate states.

Now we rewrite them in the circular-polarization basis. The four independent combinations may be taken to be

$$(a_1^+ + ib_1^+)(a_{-1}^+ + ib_{-1}^+)|0\rangle, \tag{4.21}$$
$$(a_1^+ - ib_1^+)(a_{-1}^+ - ib_{-1}^+)|0\rangle, \tag{4.22}$$
$$(a_1^+ + ib_1^+)(a_{-1}^+ - ib_{-1}^+)|0\rangle, \tag{4.23}$$
$$(a_1^+ - ib_1^+)(a_{-1}^+ + ib_{-1}^+)|0\rangle. \tag{4.24}$$

The corresponding angular momenta about the axis of motion are

$$M = +2, \qquad -2, \qquad 0, \qquad 0. \tag{4.25}$$

A compact way to display the bookkeeping is the following.

The crucial interpretation now comes exactly in the rhythm of the lecture. First: we have states with $M = \pm 2$. Could this be a spin-two particle? A *massive* spin-two particle would have five states, with $M = 2, 1, 0, -1, -2$. Those are not here. We are missing the $M = \pm 1$ pieces entirely. Therefore the only sensible reading is that the $M = \pm 2$ pair is a *massless* spin-two particle. In other words, the closed string contains the graviton.

Then comes the second step: what is left over? The answer is two independent $M = 0$ states. The lecture identifies these as two scalar particles. One linear combination is the dilaton; the other is the axion. Susskind does not derive the normalized combinations carefully on the board, so we should not pretend to do more than the lecture does. The safe statement is that the lowest level-matched closed-string sector contains

$$\text{graviton} \;+\; \text{dilaton} \;+\; \text{axion}, \tag{4.26}$$

all massless at this stage of the formal discussion.

This is the first real payoff of the closed-string spectrum. Open strings gave the photon. Closed strings, once the extra constraint is imposed, give the graviton.

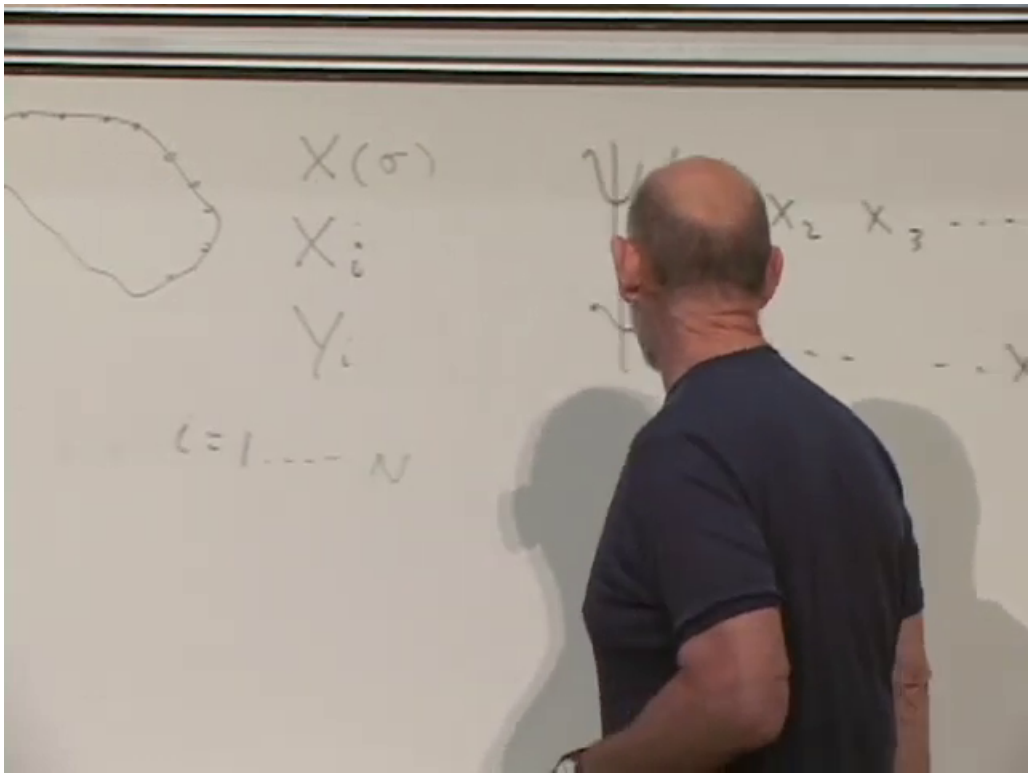

Figure 4.4: Discrete closed string and cyclic wavefunction labeling.

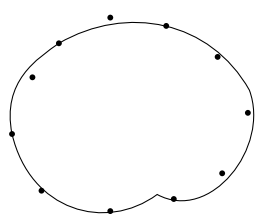

Figure 4.5: A discretized closed loop, used to make cyclic relabeling explicit.

4.7 Why Level Matching Holds: Shifting the Origin of σ

Only one major question is left hanging: why should the left-moving and right-moving energies be equal? Here the lecture returns to the marked point $\sigma = 0$ and asks whether it was ever physical to begin with.

A good place to start is the discrete string. Think of the loop as a large collection of points. Then instead of $X(\sigma)$ and $Y(\sigma)$, one writes

$$X(\sigma), \qquad X_i, \qquad Y_i, \qquad i = 1, \ldots, N. \tag{4.27}$$

The board layout for this step is worth keeping because it shows the discrete loop, the notation, and the wavefunction block together:

A clean schematic of the discretized loop is useful nearby:

To keep the notation from becoming unbearable, the lecture temporarily suppresses the Y_i variables

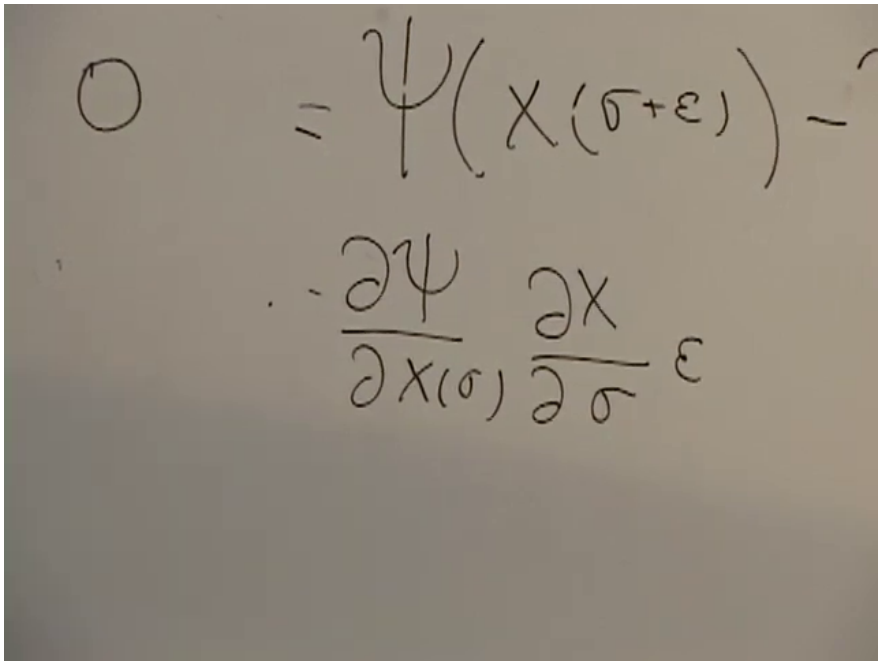

Figure 4.6: Small-ϵ shift of the wavefunctional argument.

and speaks only about the x-coordinates. The quantum state is then a wavefunction

$$\Psi(x_1, x_2, \ldots, x_N). \tag{4.28}$$

But if the point called $\sigma = 0$ is not special, why should the description begin with x_1 rather than x_2? The natural requirement is not full symmetry under arbitrary reorderings, which would rip the string open, but invariance under cyclic relabeling:

$$\Psi(x_1, x_2, \ldots, x_N) = \Psi(x_2, \ldots, x_N, x_1) = \Psi(x_3, \ldots, x_N, x_1, x_2). \tag{4.29}$$

In discrete notation the shift is simply

$$X_i \to X_{i+1}. \tag{4.30}$$

This is the real content of "there is no preferred point on the string."

Now pass to the continuum. The discrete cyclic shift becomes

$$\sigma \to \sigma + \epsilon. \tag{4.31}$$

The wavefunction is now a functional of the whole profile $X(\sigma)$. In cleaned form, the board statement is

$$0 = \Psi\big(X(\sigma + \epsilon)\big) - \Psi\big(X(\sigma)\big), \tag{4.32}$$

with the understanding that the upper line in the blackboard image is cropped and this subtraction is reconstructed from the surrounding transcript. The lecture then zooms in on the first-order variation, and that local structure is visible in the board close-up:

At the board level the infinitesimal change is written as

$$\delta\Psi \sim \frac{\partial \Psi}{\partial X(\sigma)} \frac{\partial X}{\partial \sigma}\, \epsilon. \tag{4.33}$$

This is faithful to what is written, but we must immediately say what the lecture itself then says: this is not a separate equation for one special value of σ. The shift moves *every* point on the string.

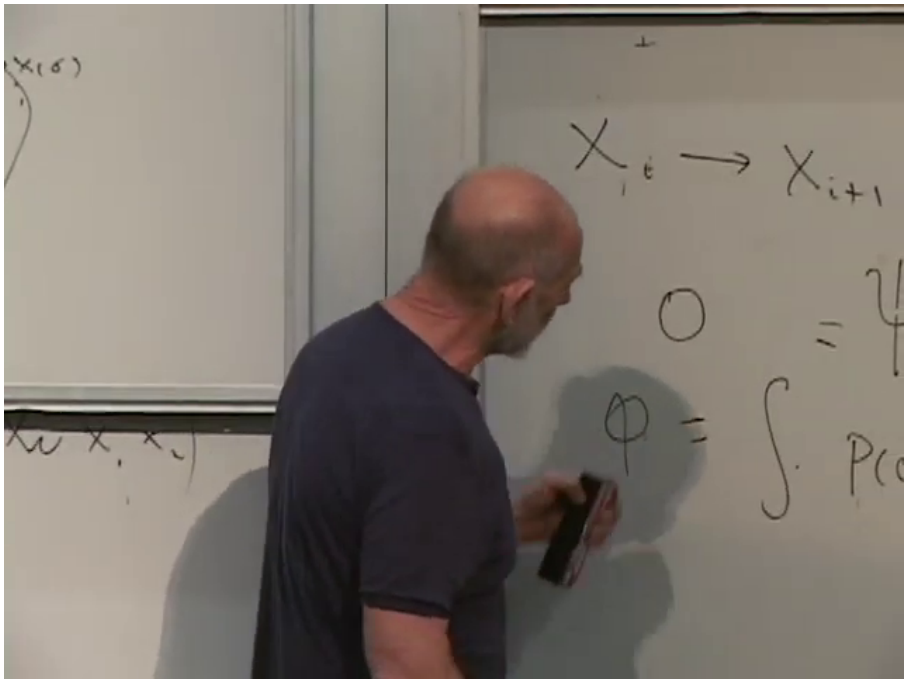

Figure 4.7: Discrete shift and the emerging reparameterization constraint.

4.7.1 Question & Answer

Question. Is $\sigma = 0$ really a physically special point, and is the variation equation separate for each σ?

Answer. No on both counts. If no point on the loop is physically preferred, then shifting the origin of σ must leave the state unchanged. And since the shift changes the coordinate of every point on the string, one must add the variations from all σ-points. In the continuum that means integrating:

$$\int d\sigma \, \frac{\delta \Psi}{\delta X(\sigma)} \, \partial_\sigma X(\sigma) = 0. \tag{4.34}$$

Here we have upgraded the board's ordinary derivative notation to the appropriate functional derivative notation, but the logic is exactly the lecture's logic.

Now we return to elementary quantum mechanics. Differentiation of the wavefunction with respect to a coordinate is the action of the corresponding momentum operator. Suppressing $\hbar$,

$$P(\sigma) \sim -i \, \frac{\delta}{\delta X(\sigma)}. \tag{4.35}$$

Therefore the invariance condition becomes

$$\int d\sigma \, P(\sigma) \, \partial_\sigma X(\sigma) = 0. \tag{4.36}$$

The board transition into this formula is partly preserved in the next frame:

In the simple nonrelativistic string model used throughout these lectures, momentum is just velocity, so

$$P(\sigma) = \partial_\tau X(\sigma). \tag{4.37}$$

Hence the constraint becomes

$$\int d\sigma\, \partial_\tau X\, \partial_\sigma X = 0. \tag{4.38}$$

This is already the essential result. To see why it is level matching, compare it with the left-right decomposition of the energy. Suppressing the duplicated transverse directions for a moment, the difference of the two energy densities is

$$\frac{1}{2}\Big[(\partial_\tau X + \partial_\sigma X)^2 - (\partial_\tau X - \partial_\sigma X)^2\Big] = 2\,\partial_\tau X\, \partial_\sigma X. \tag{4.39}$$

So, up to the same harmless normalization ambiguity that Susskind himself flags on the board,

$$E_L - E_R \propto \int d\sigma\, \partial_\tau X\, \partial_\sigma X. \tag{4.40}$$

But the integral on the right must vanish if the theory is invariant under shifting the origin of σ. Therefore

$$E_L = E_R. \tag{4.41}$$

This is the beautiful fact the lecture is aiming at. Level matching is not an arbitrary cleanup rule imposed after the fact. It is the statement that the closed string has no physically distinguished point. And the rule throws away exactly the large chunk of the naïve spectrum that refused to organize itself into sensible particle multiplets.

4.8 Clarifications, Fundamentality, and the M-Theory Turn

The lecture closes by cleaning up one possible confusion and then widening the horizon.

The confusion is linguistic. Susskind has used the words “left” and “right” in two different senses. One sense refers to intrinsic propagation along the string: motion toward increasing σ or decreasing σ. The other sense refers to circular polarization in ordinary x-y space, as when one speaks of right-handed or left-handed photon polarization. These are completely different notions. The first has to do with the sign of n; the second has to do with combinations like $a \pm ib$. The lecture returns to this point explicitly, and the notes should do the same.

Then comes the philosophical coda. After all the formal machinery of open and closed strings, one may ask whether strings are truly fundamental. Susskind answers this by backing away from the word itself. The better question is not which object is ultimately fundamental in some absolute metaphysical sense, but which description is useful in a given regime of parameters. He illustrates the point with electric charges and magnetic monopoles: in one range of coupling constants the electric variables are simple and the magnetic variables are heavy and complicated; in another range the roles can reverse. The same moral carries over to string theory. There are regimes in which it is useful to think of strings as the fundamental objects, and other regimes in which D-branes become the simpler degrees of freedom and strings look more composite.

That is the lesson the lecture wants to leave us with before the next turn. Physics does not seem to be marching toward one final smallest ingredient in any naïve reductionist sense. Rather, the useful elementary objects can change as the parameters of the theory change. The next subject will be M-theory, where exactly that kind of change of viewpoint becomes indispensable.

4.8.1 Summary

The lecture begins with Noether's theorem because a symmetry argument is waiting in the wings. We then draw a closed string, mark a point $\sigma = 0$, and insist that the mark is only conventional. Periodicity leads to Fourier modes with positive and negative integers, and the closed-string energy naturally splits into left-moving and right-moving pieces. That split doubles the oscillator structure.

A naïve first look at the spectrum suggests a doubled photon story, but the lecture deliberately interrupts that guess. The correct rule is level matching: left-moving and right-moving energies must be equal. Once that rule is imposed, the first surviving closed-string excitations assemble into a massless spin-two particle together with two scalar states. The graviton appears, and beside it the dilaton and axion.

Finally, the deferred explanation is supplied. Level matching follows from invariance under shifting the origin of the σ-coordinate. A closed string with no preferred point on the loop must satisfy a momentum constraint, and that constraint is precisely the vanishing of the difference between left-moving and right-moving energies. In this way the geometry of the closed loop, the symmetry of its parameterization, and the structure of the spectrum all lock together.